CS304 MID TERM FILE SOLVED BY IQRA AZHAR

CS304

Mid term

2023

Solved by

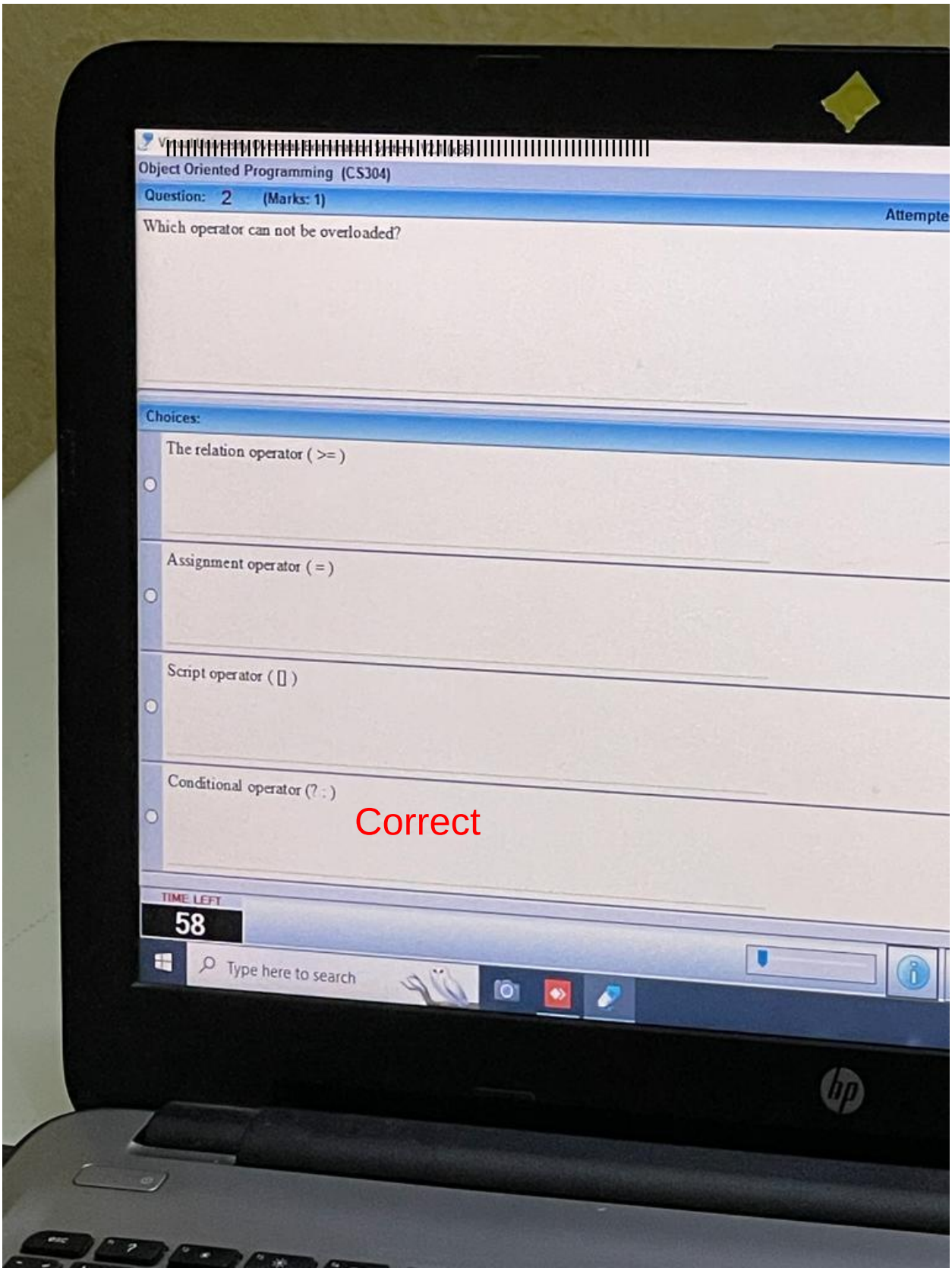
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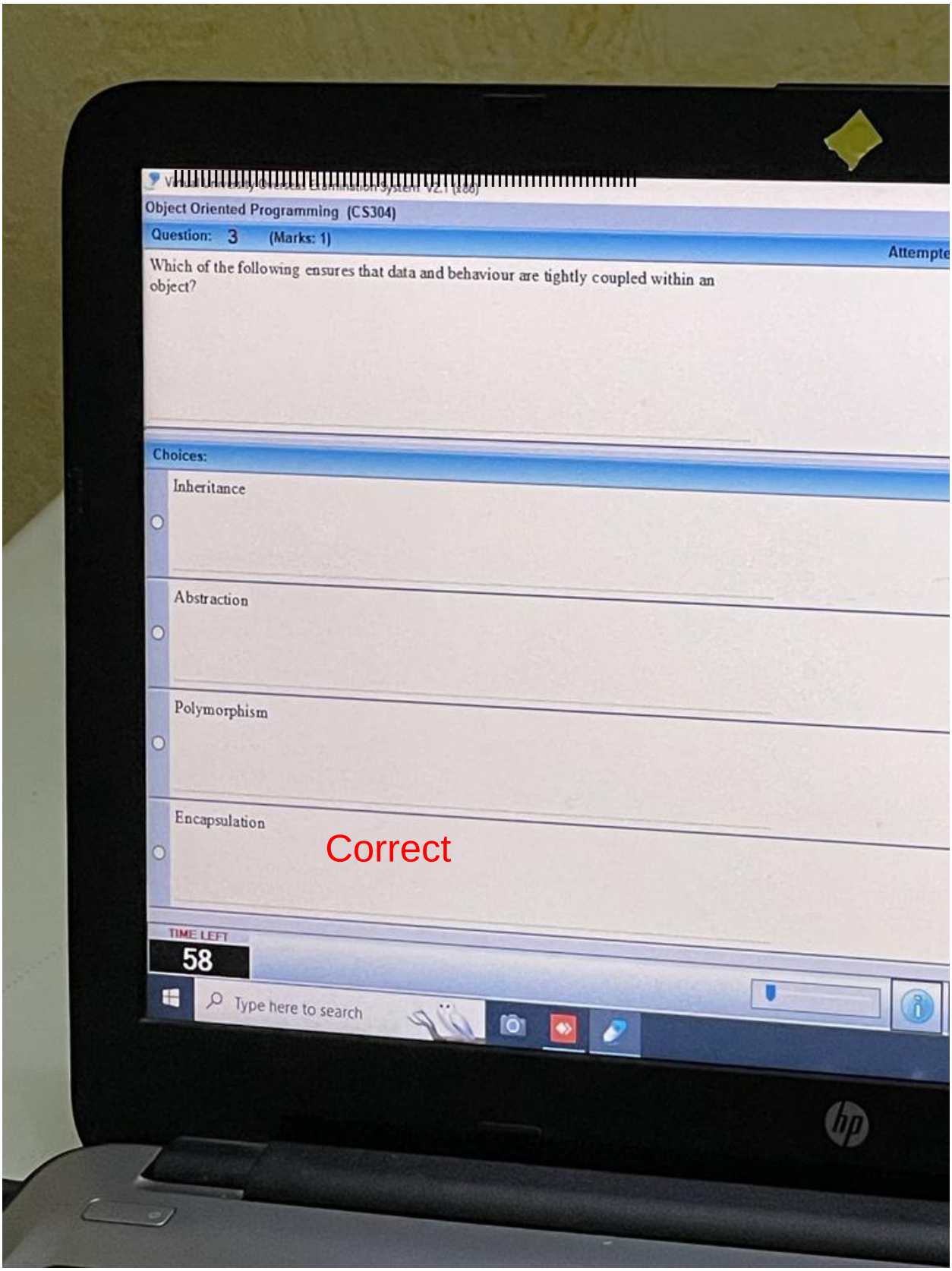
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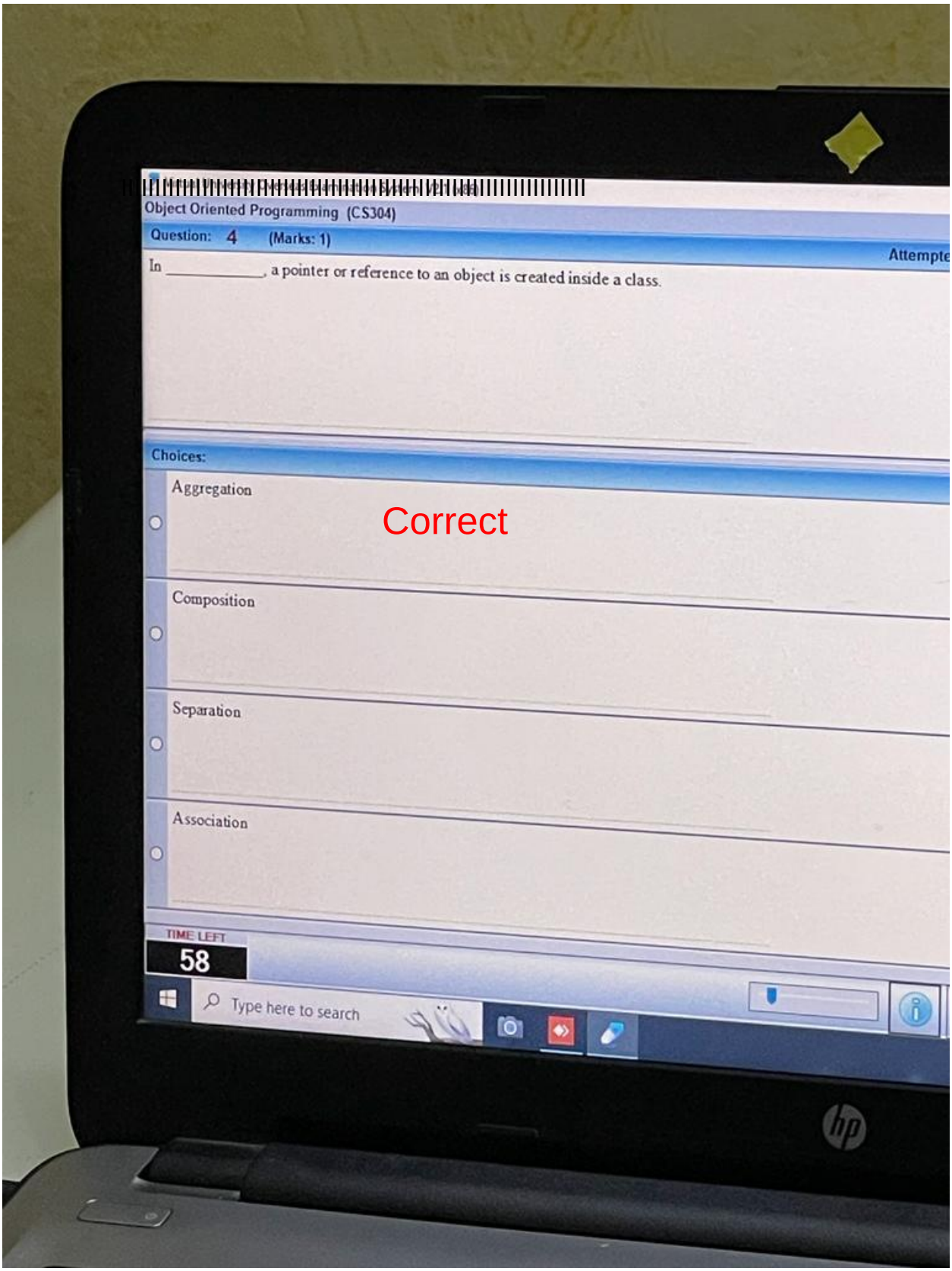
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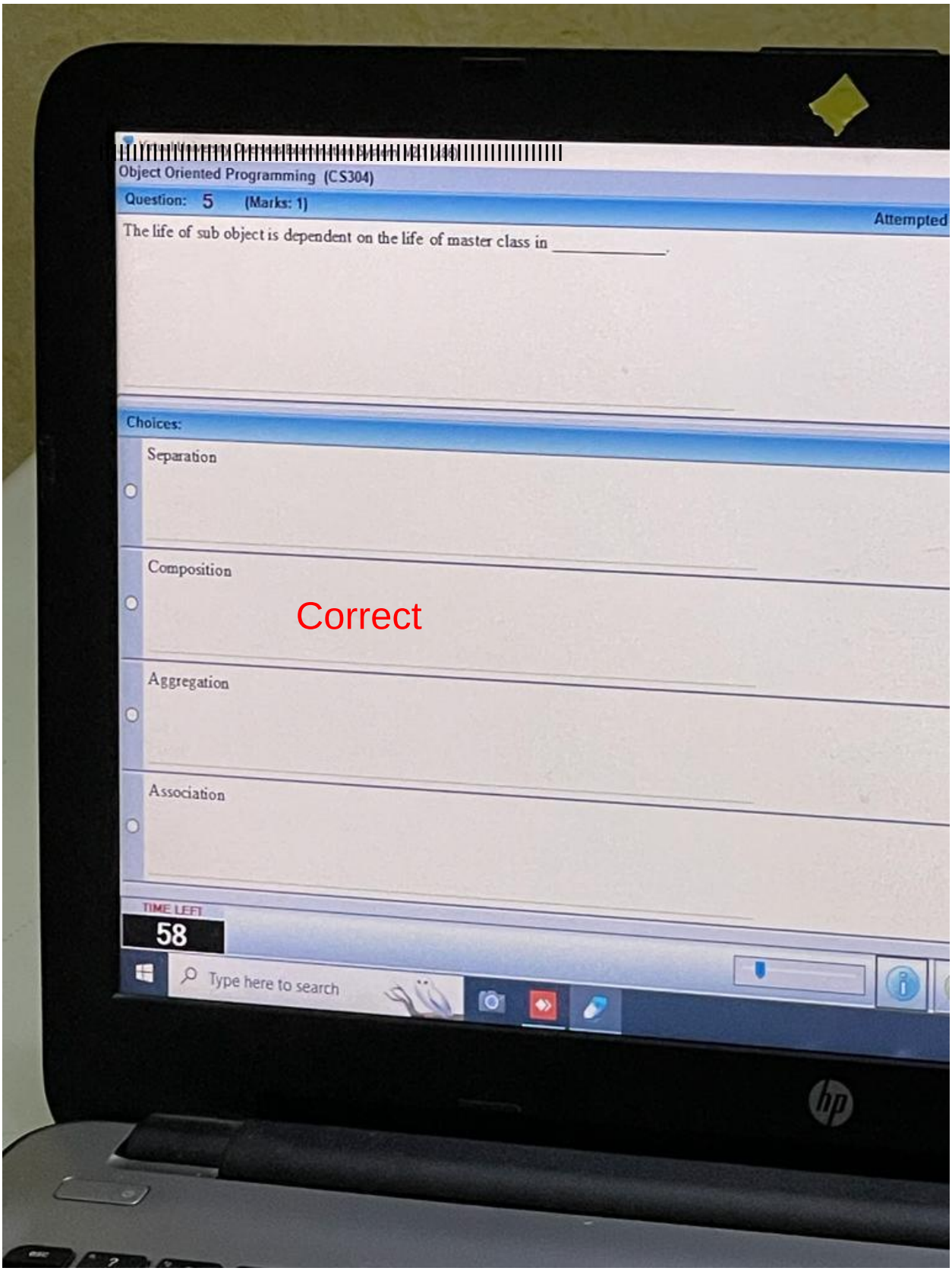
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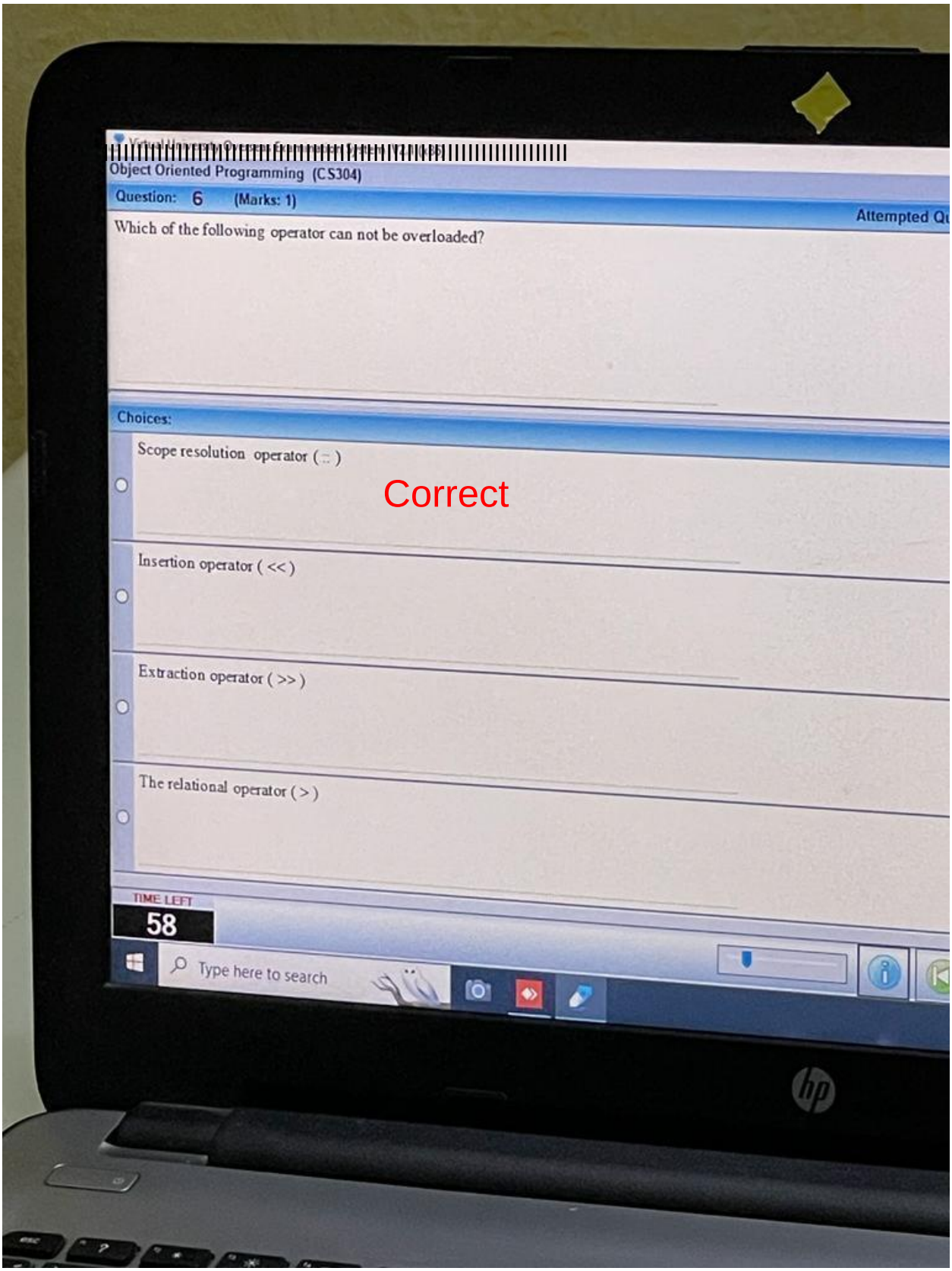
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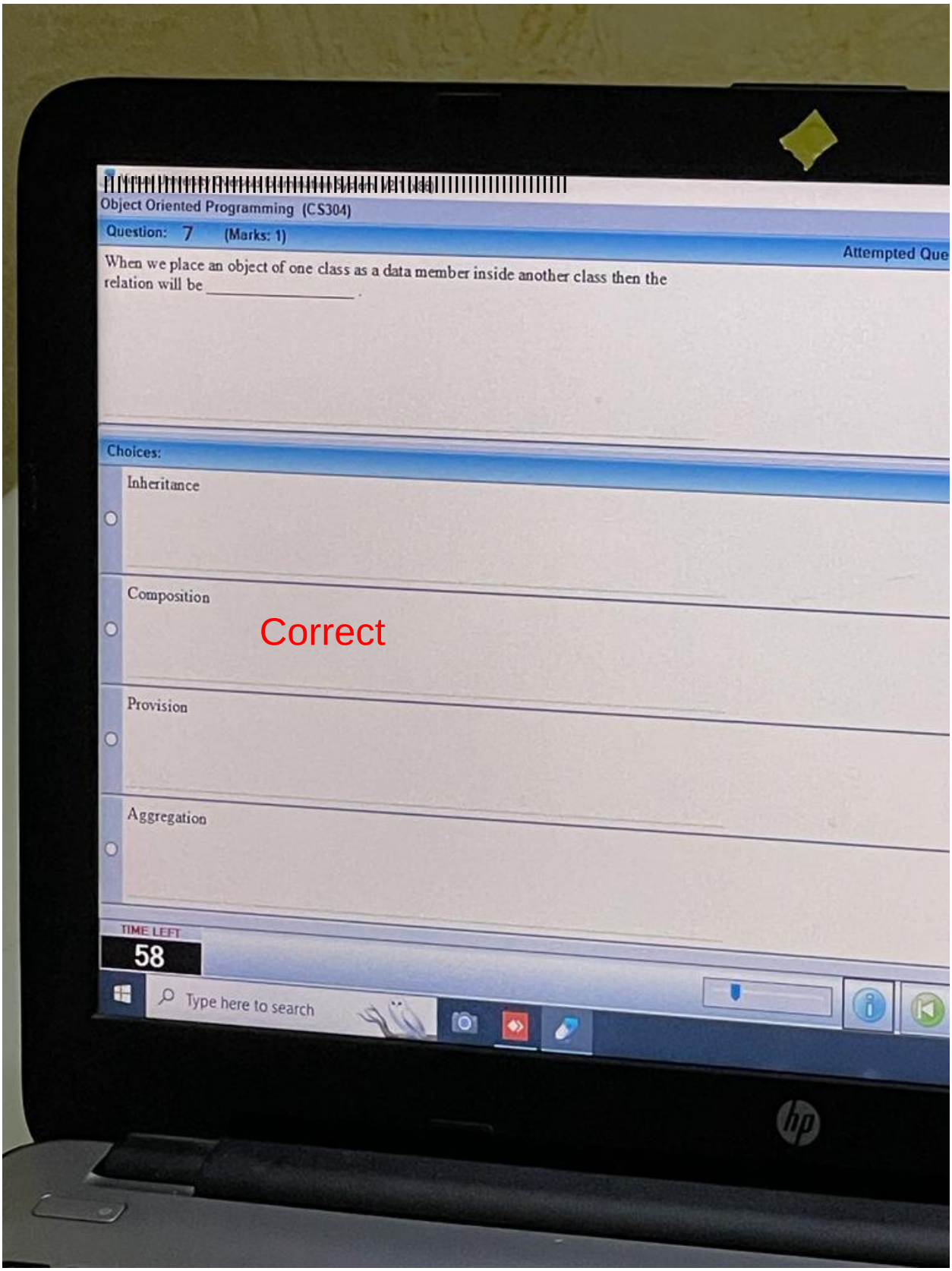
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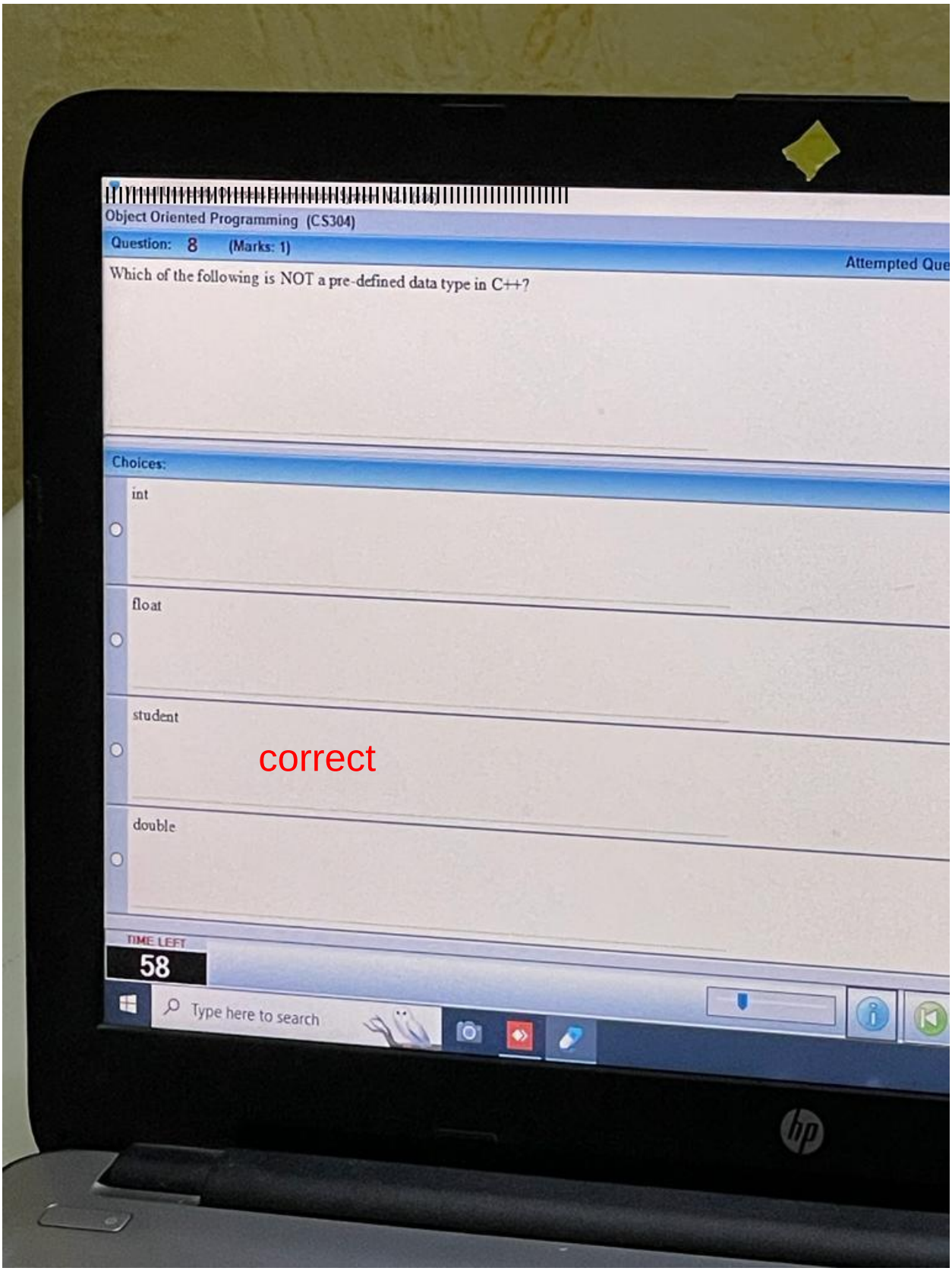
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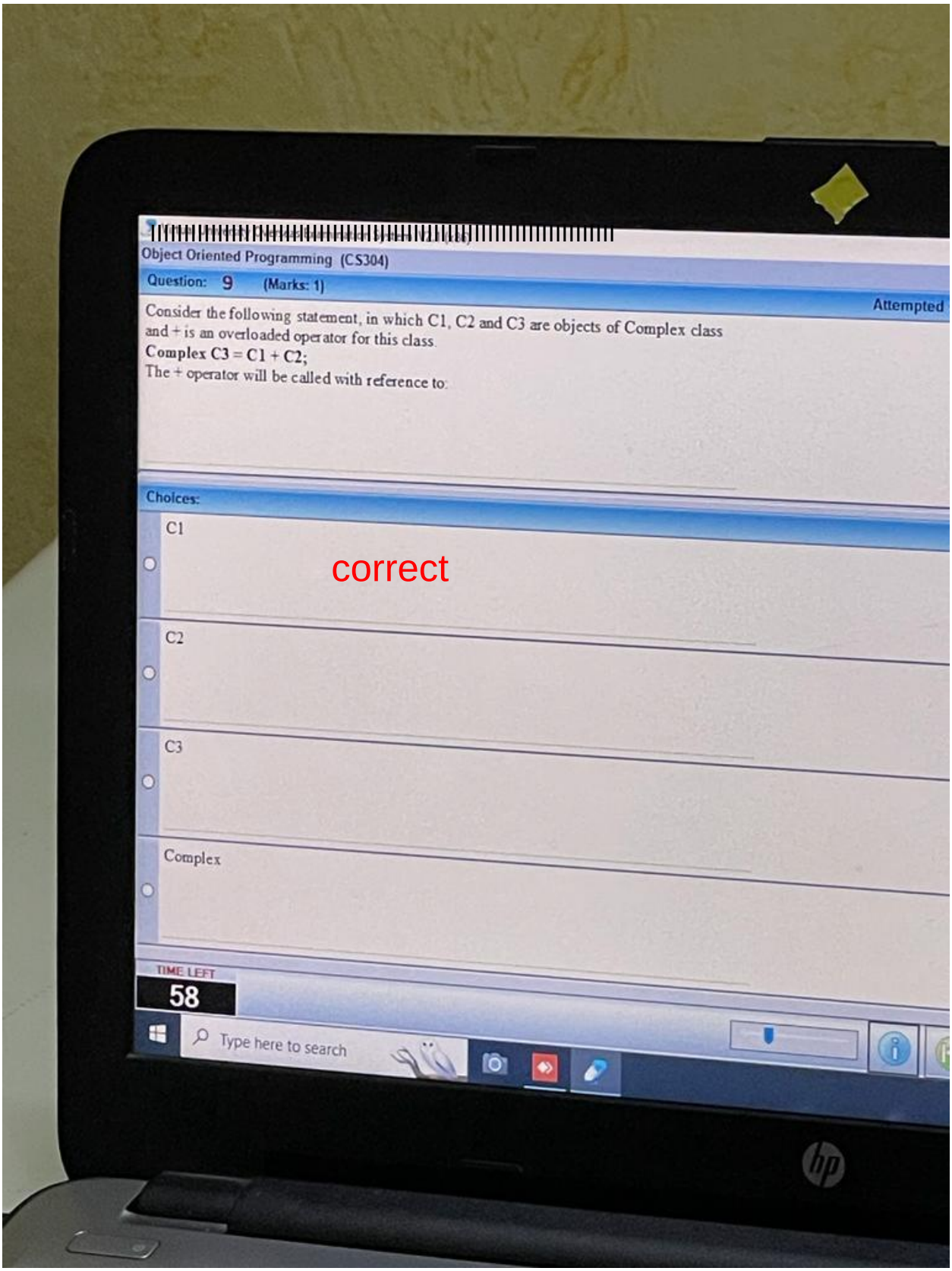
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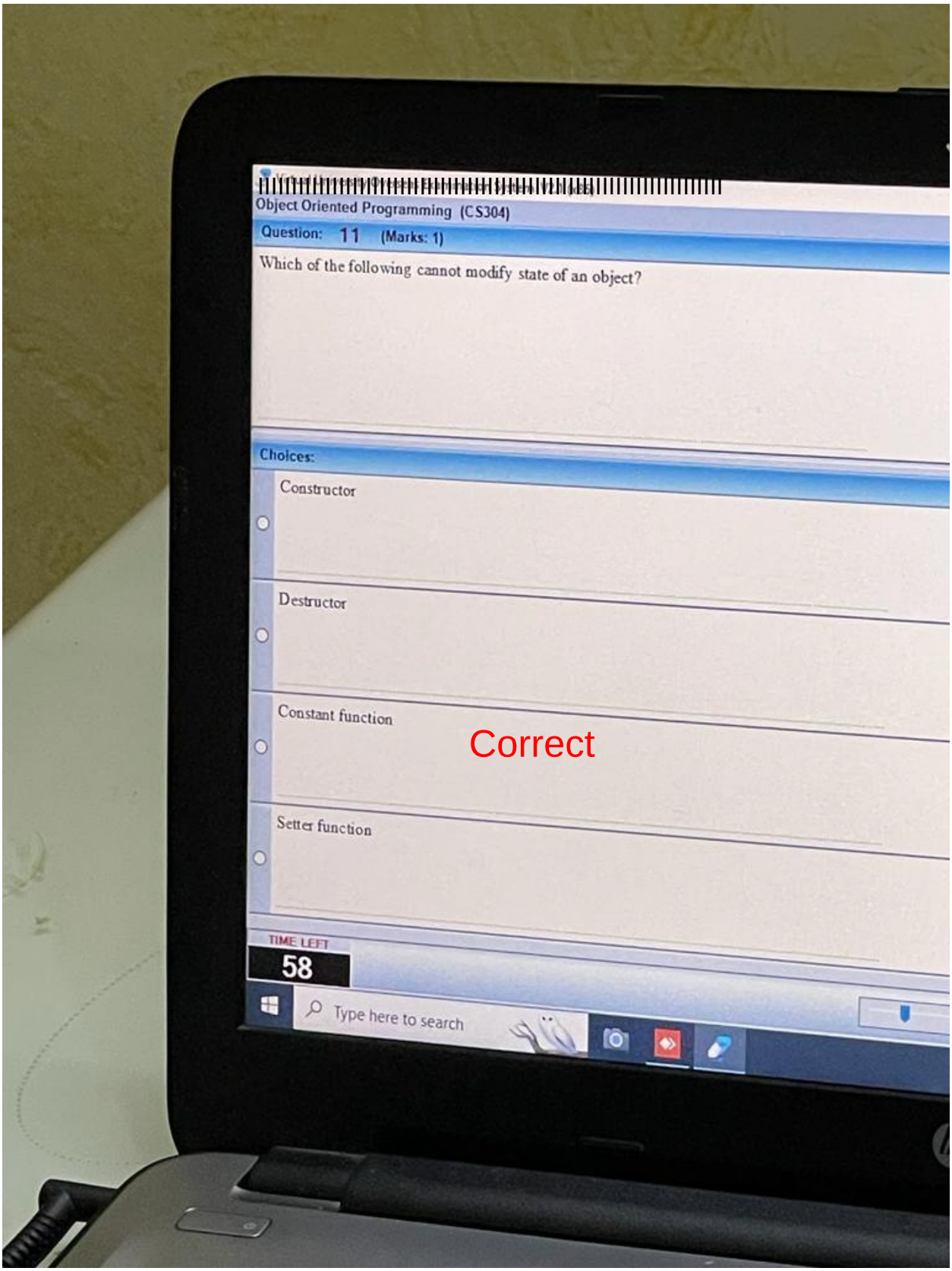
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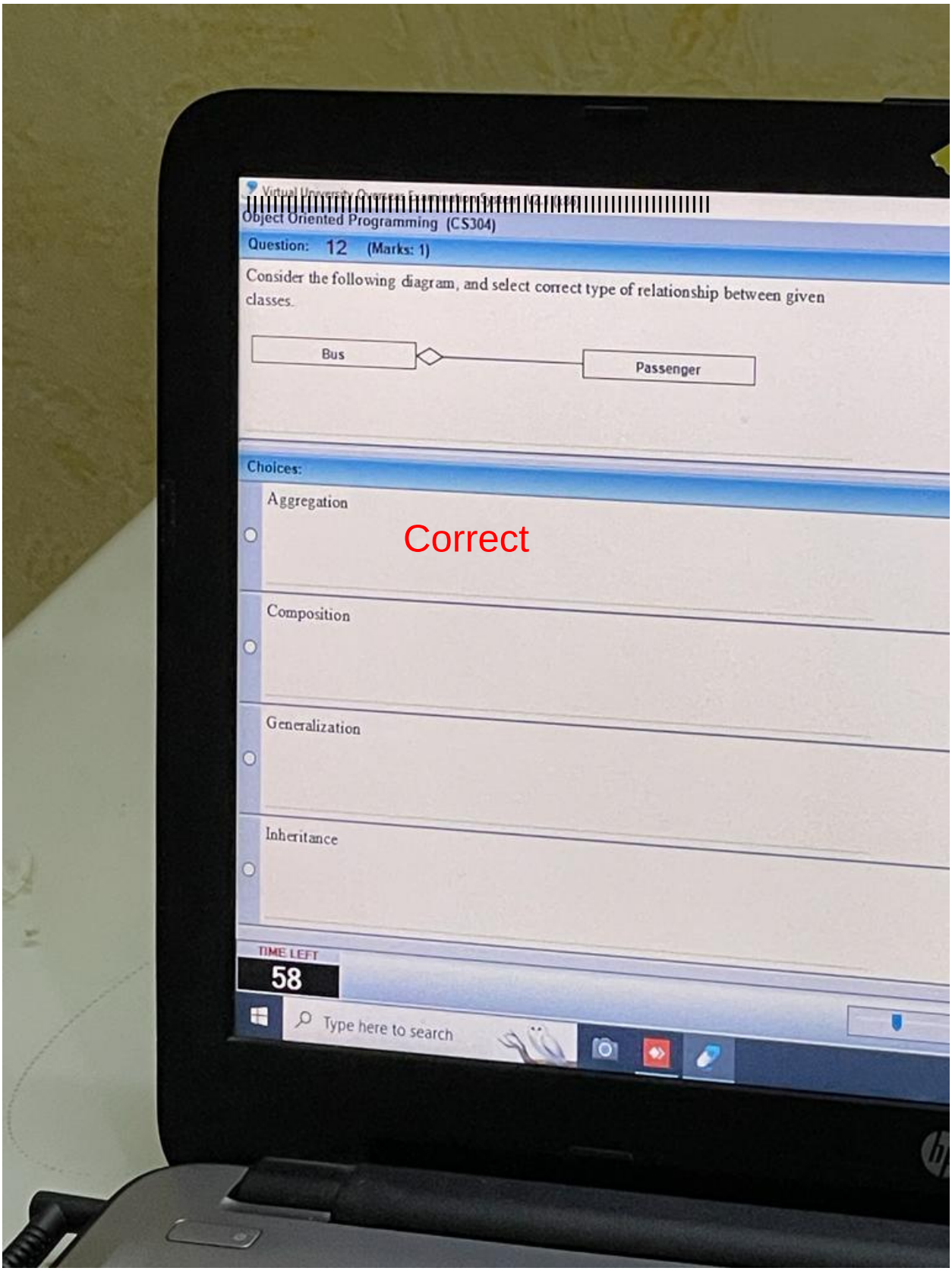
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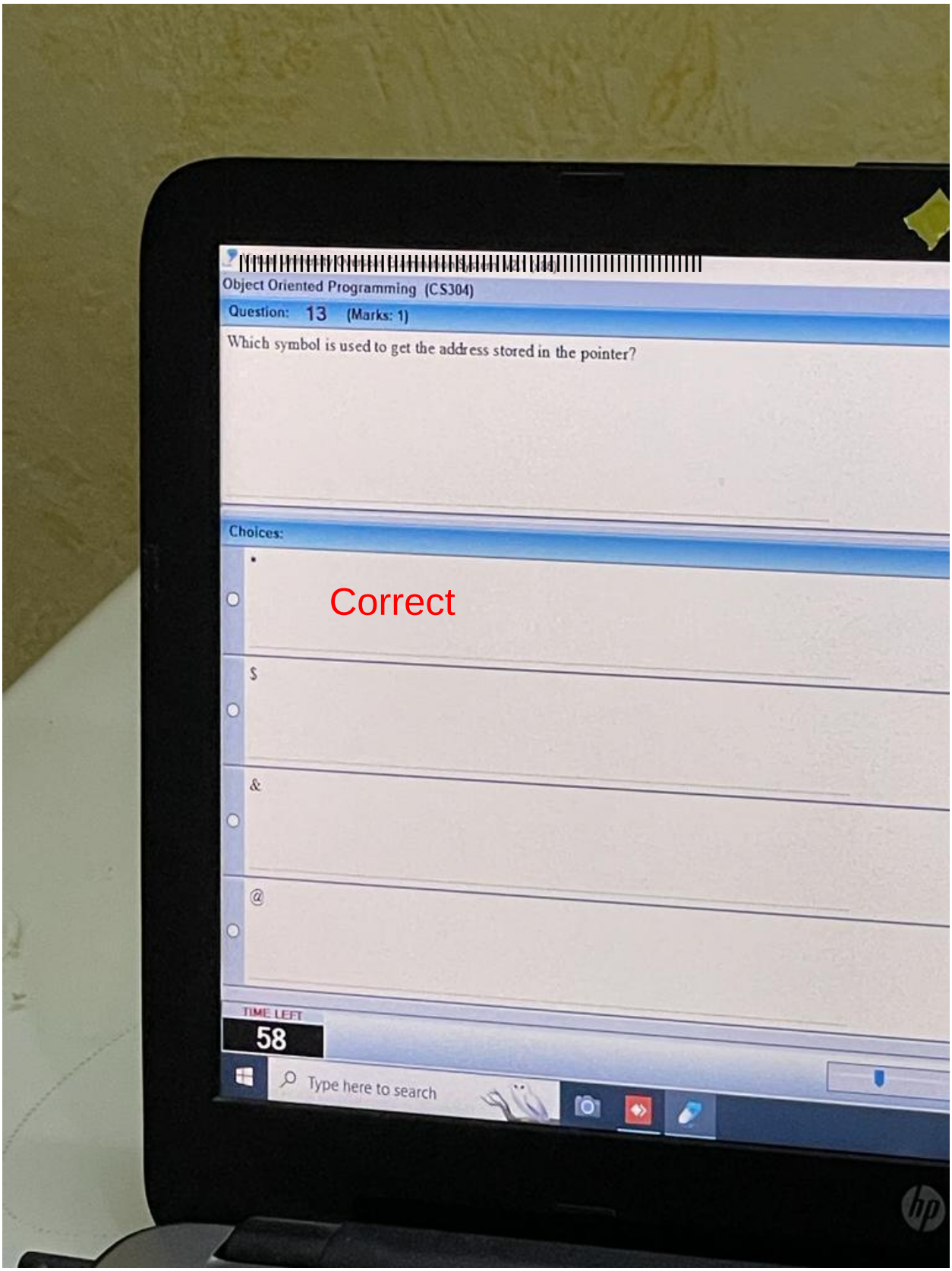
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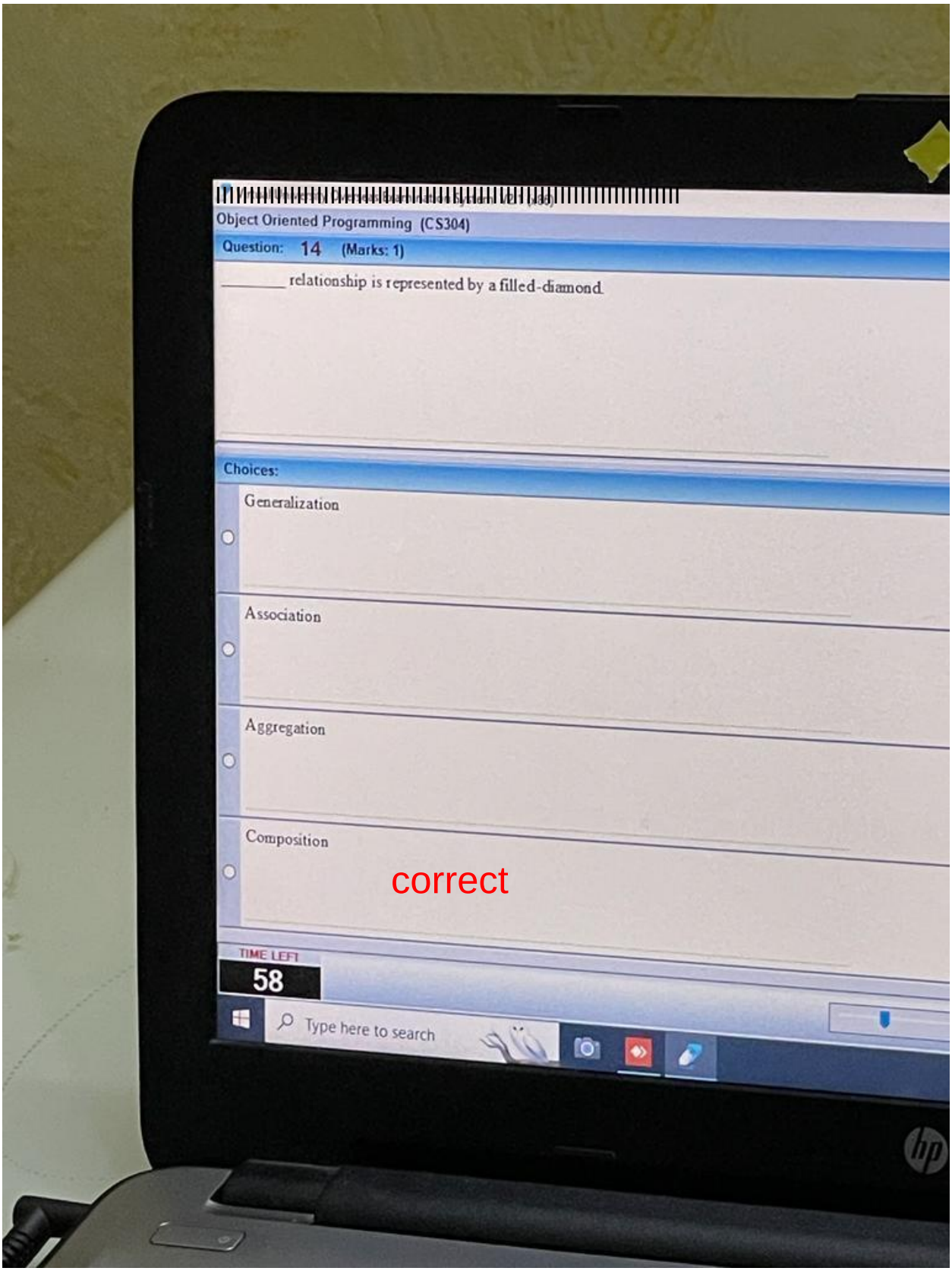
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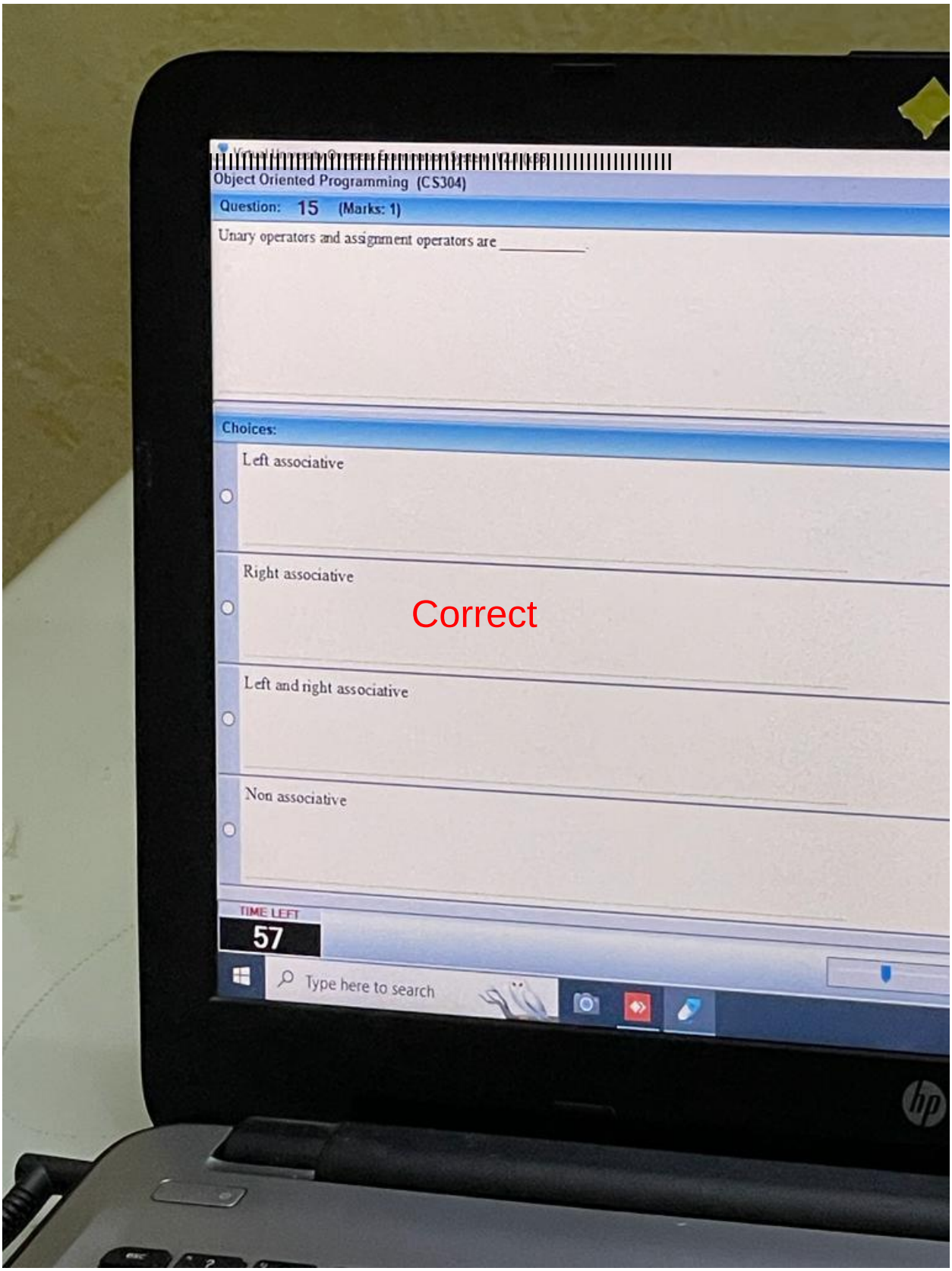
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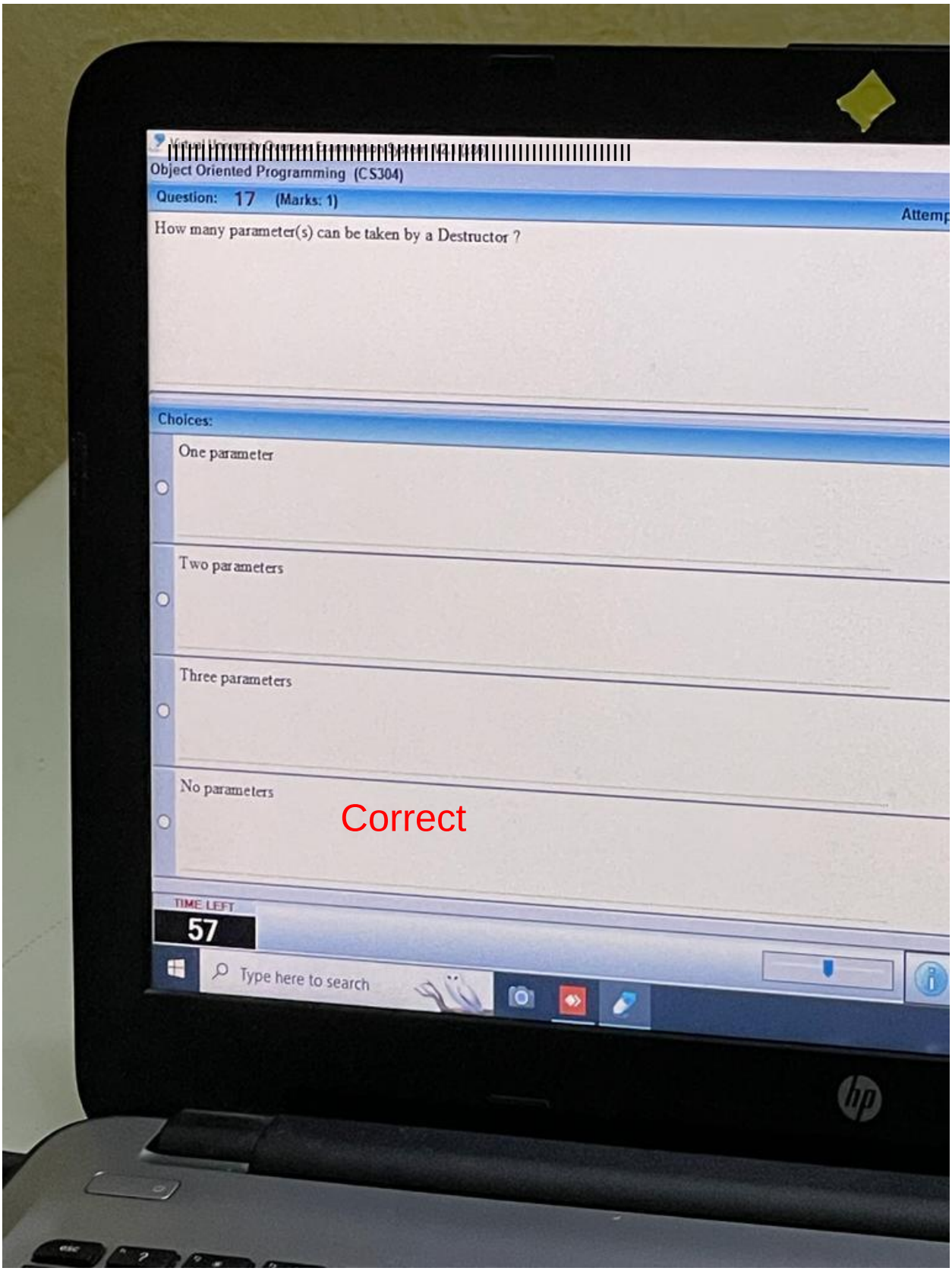
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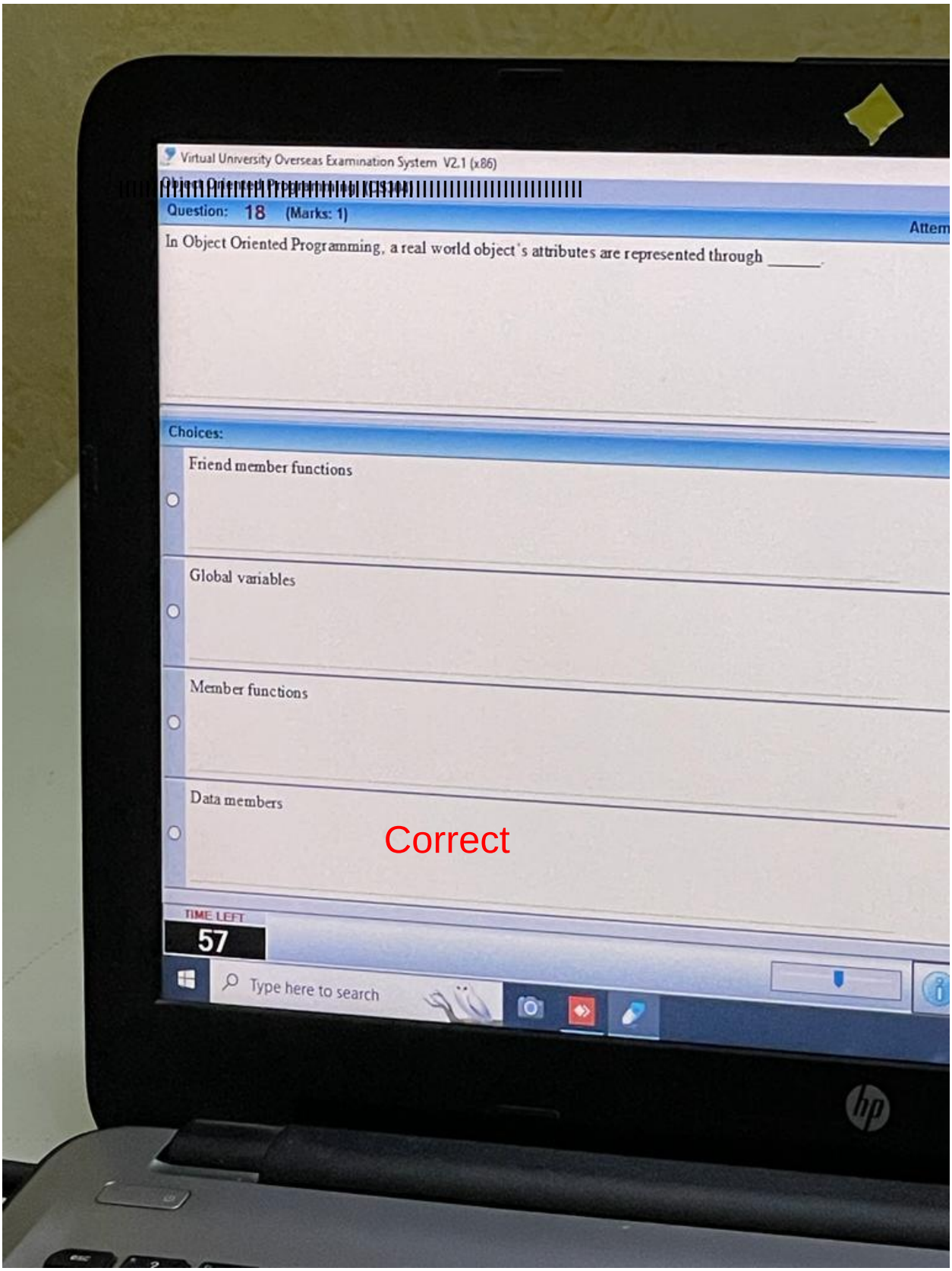
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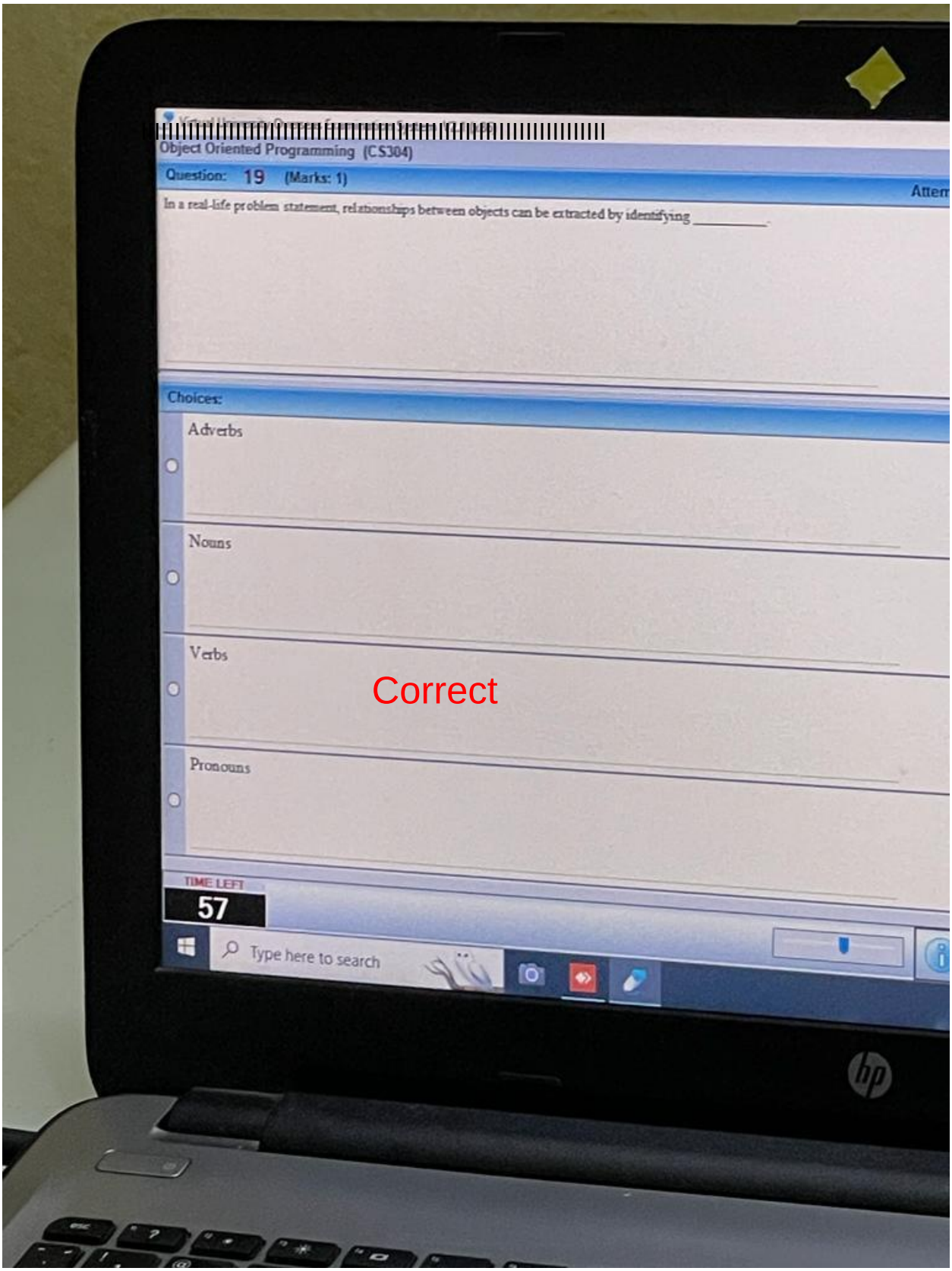
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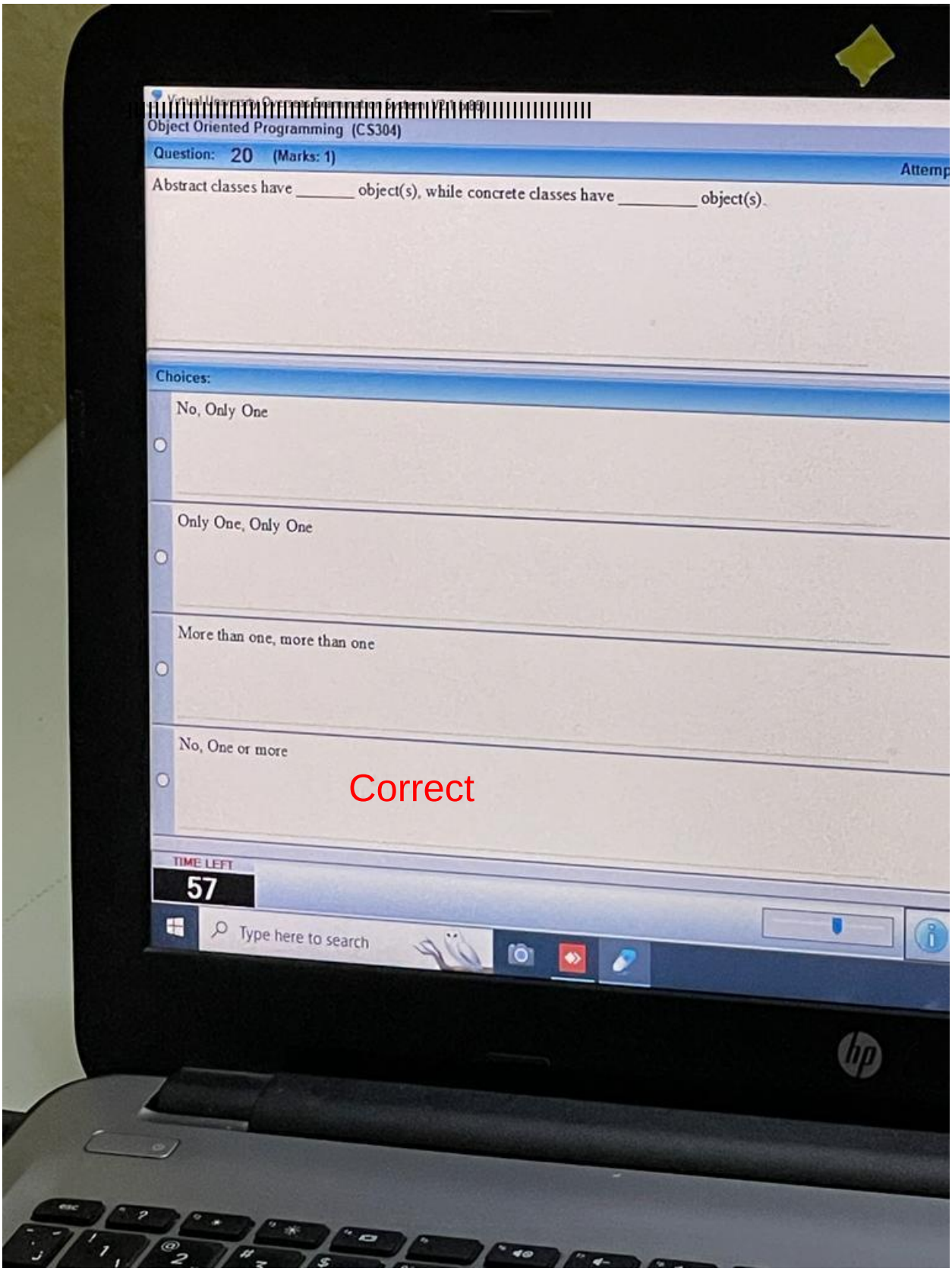
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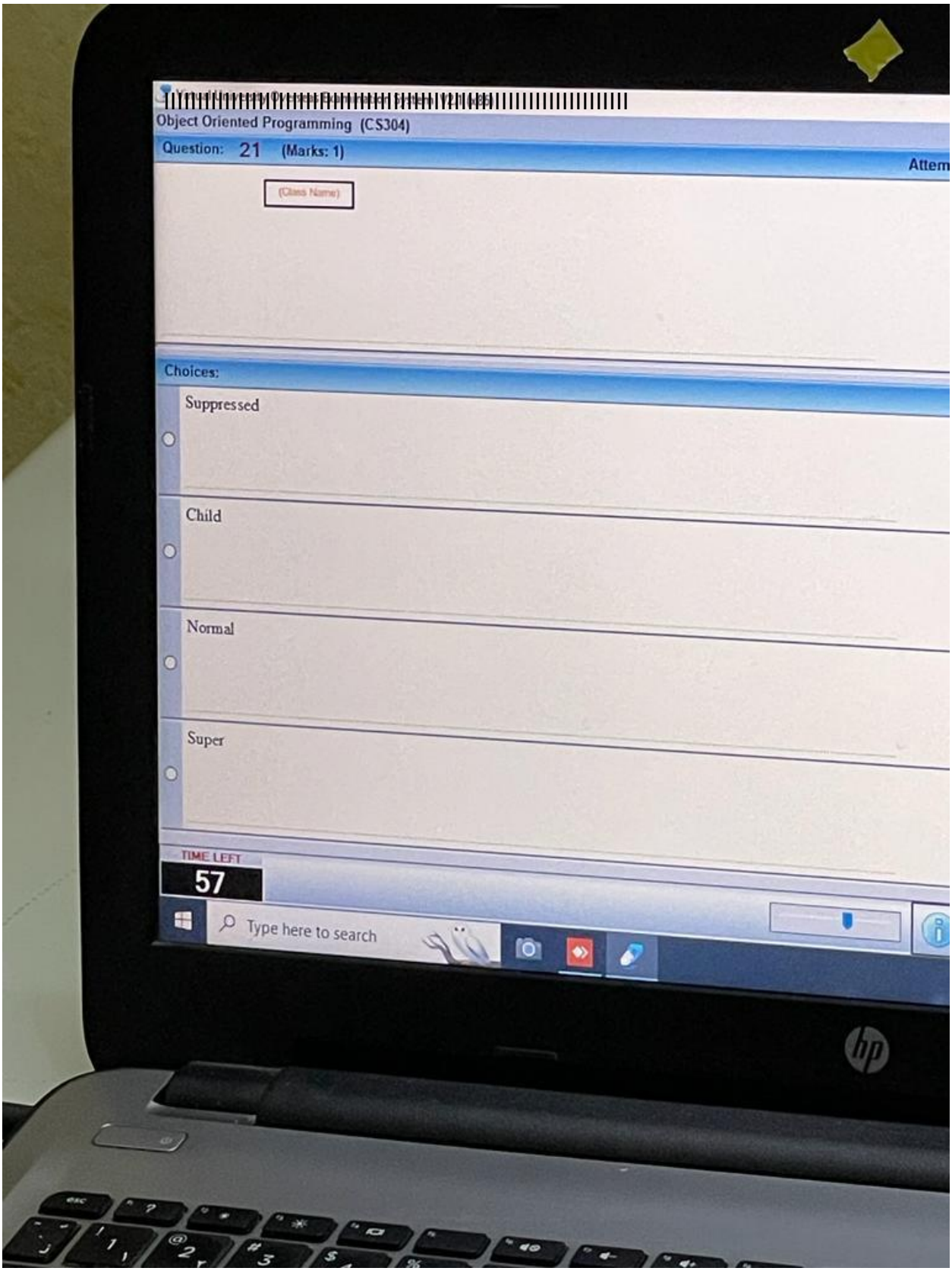
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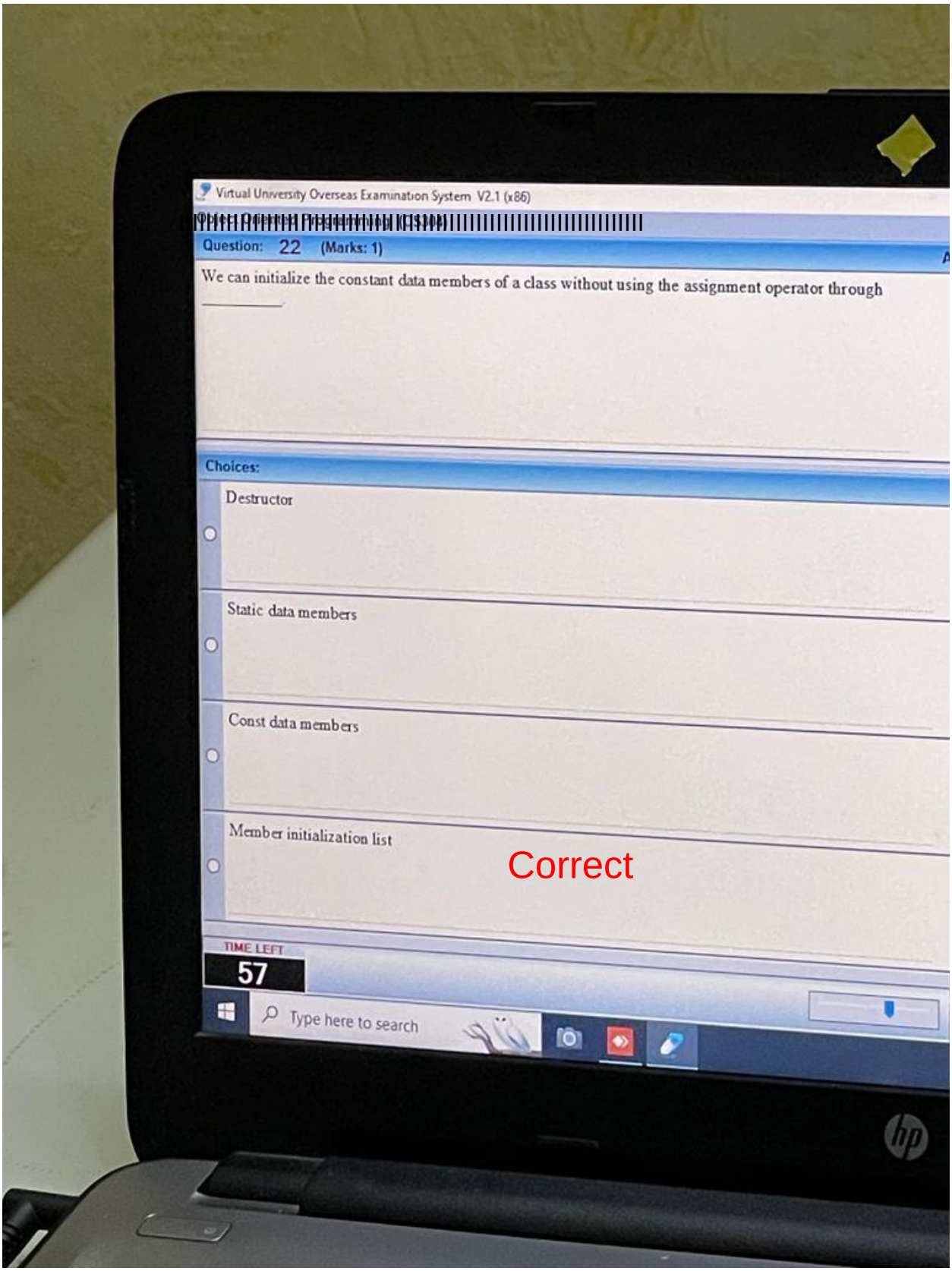
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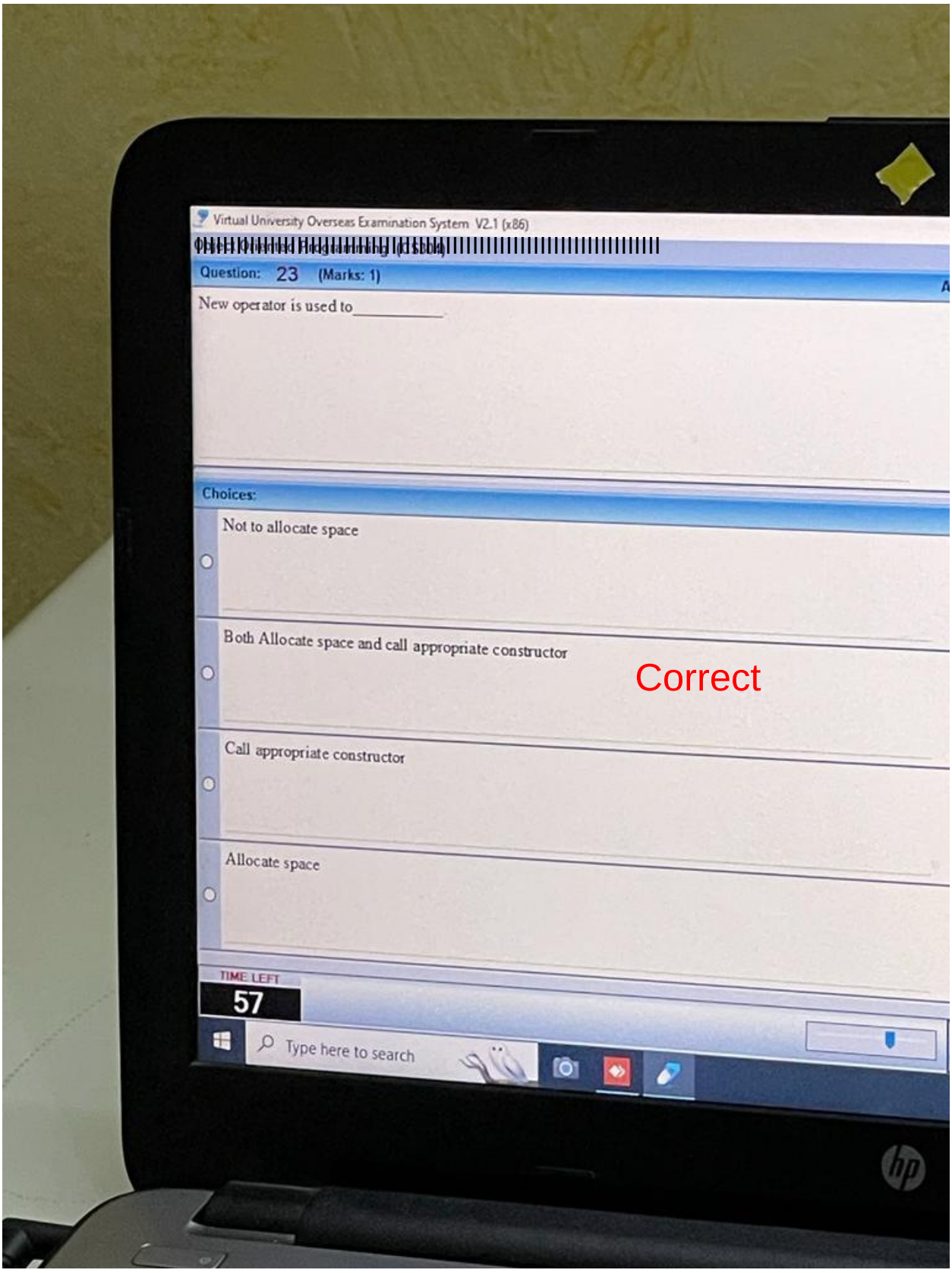
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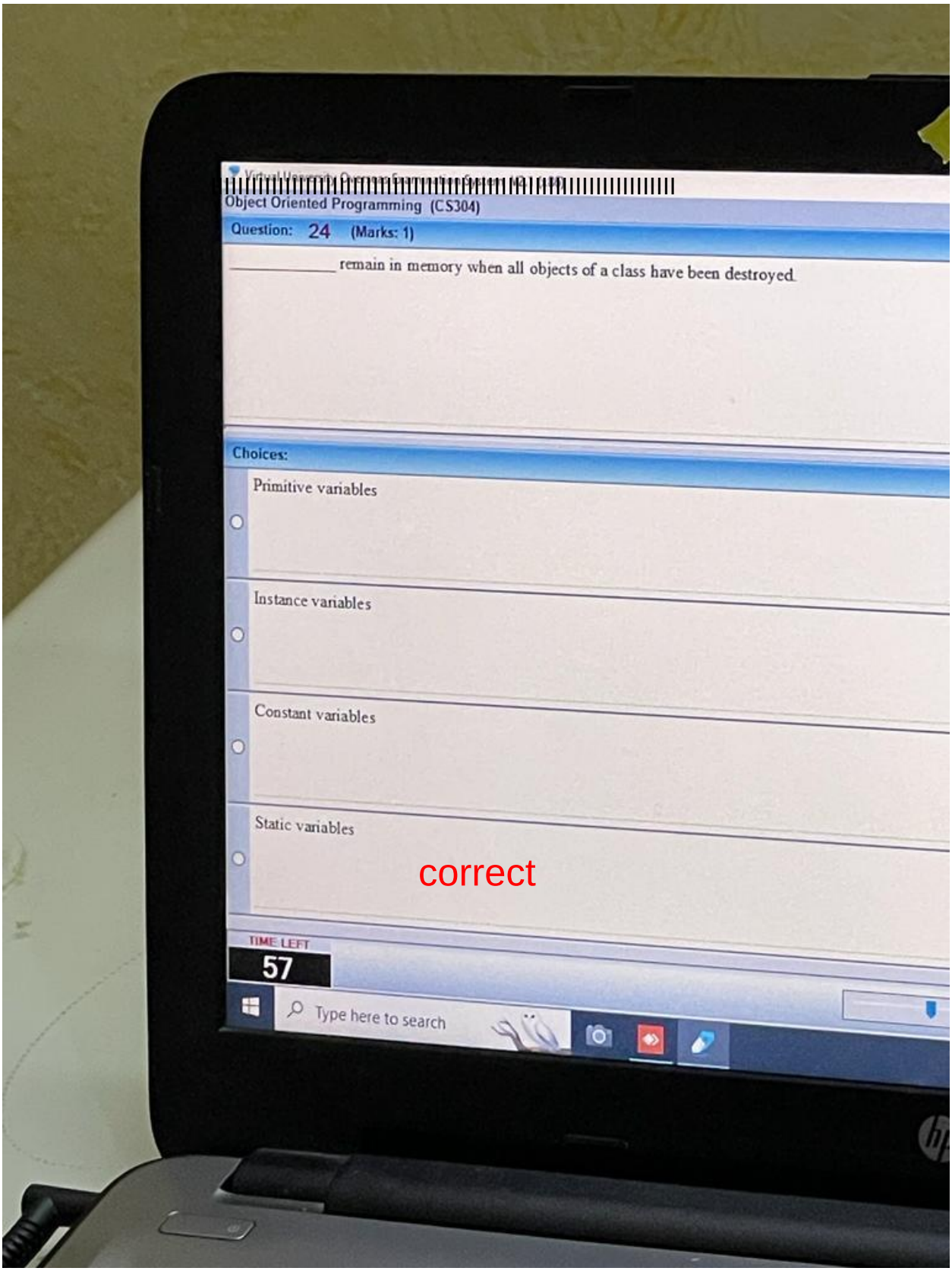
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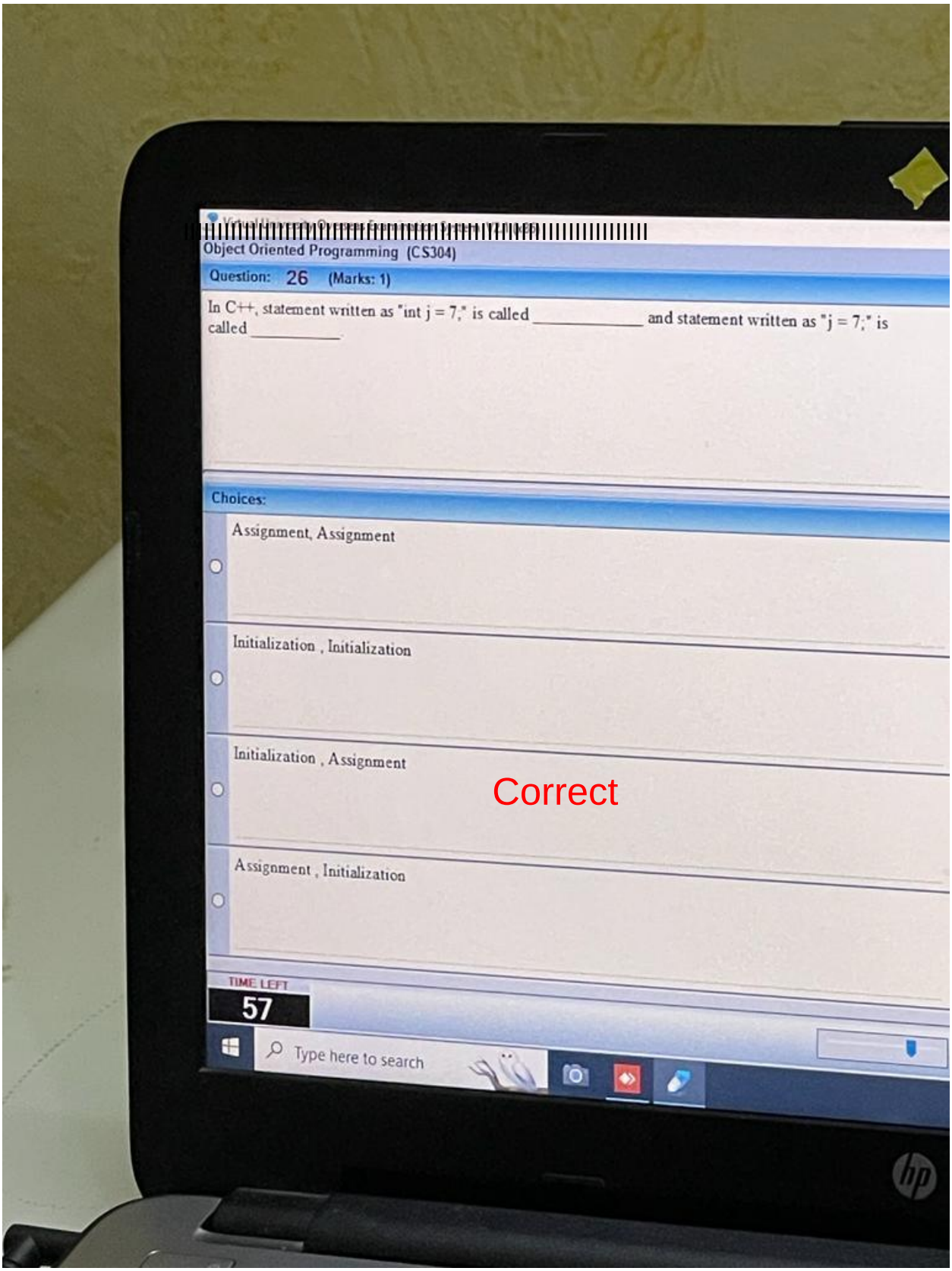
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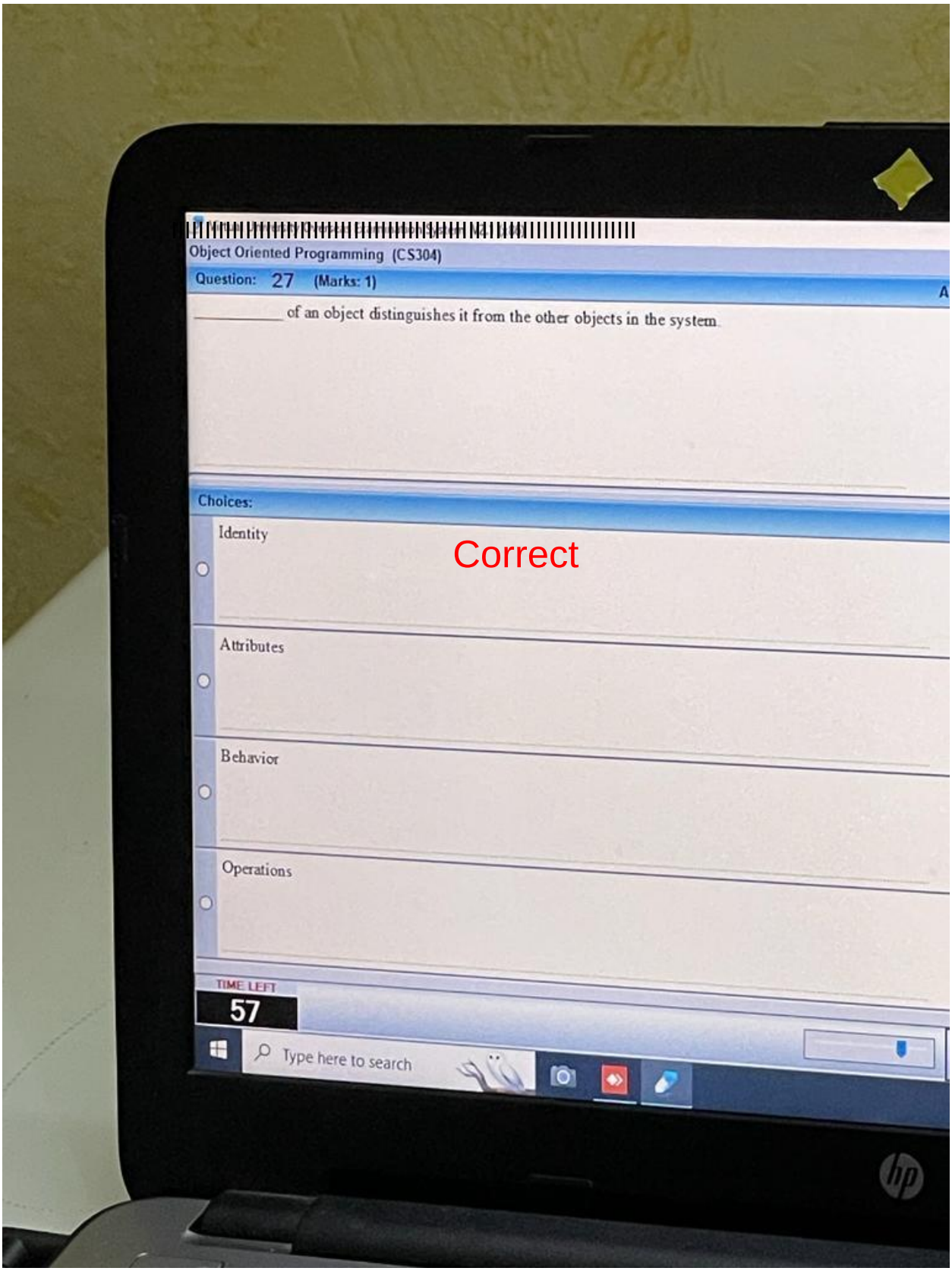
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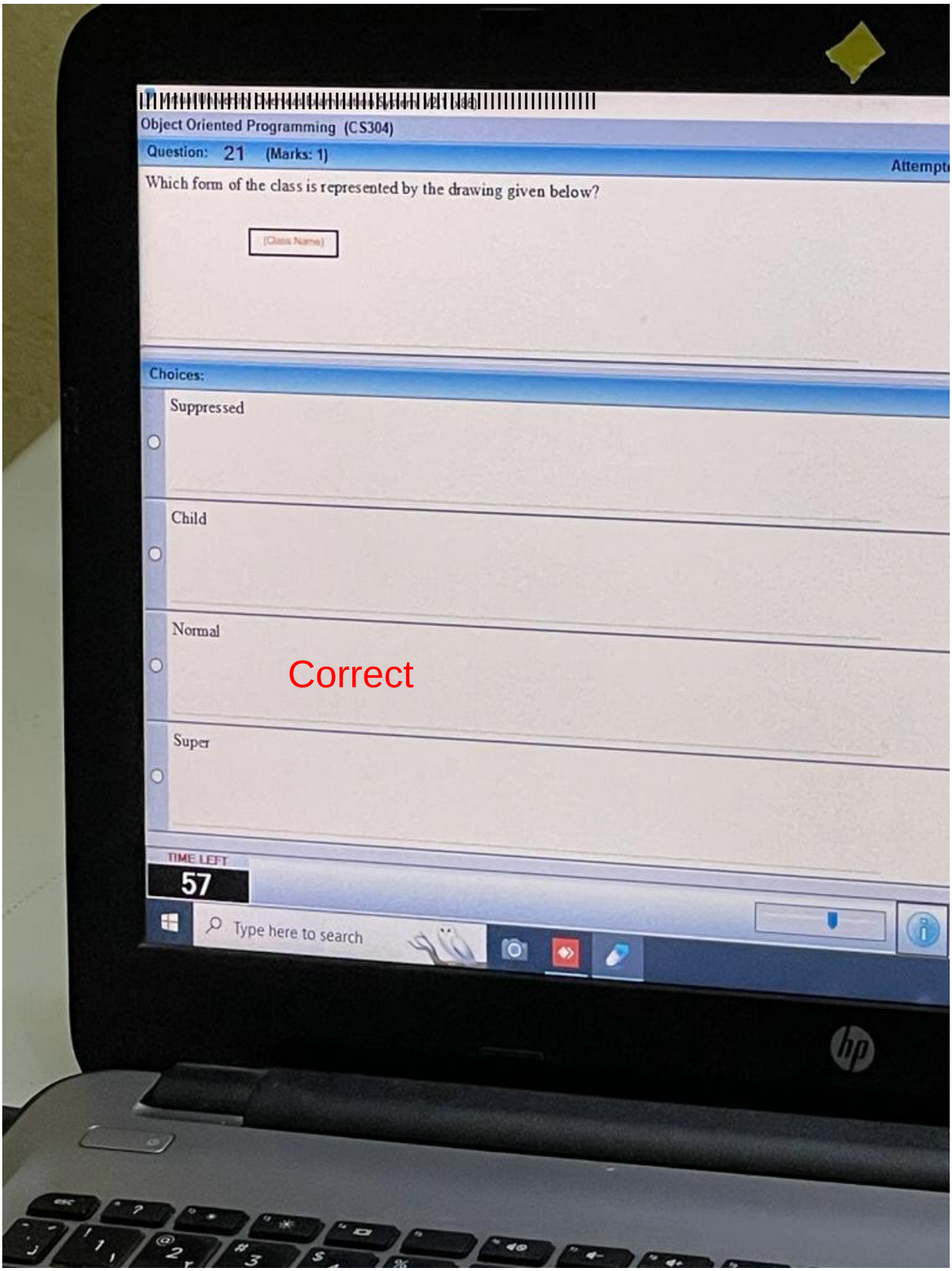
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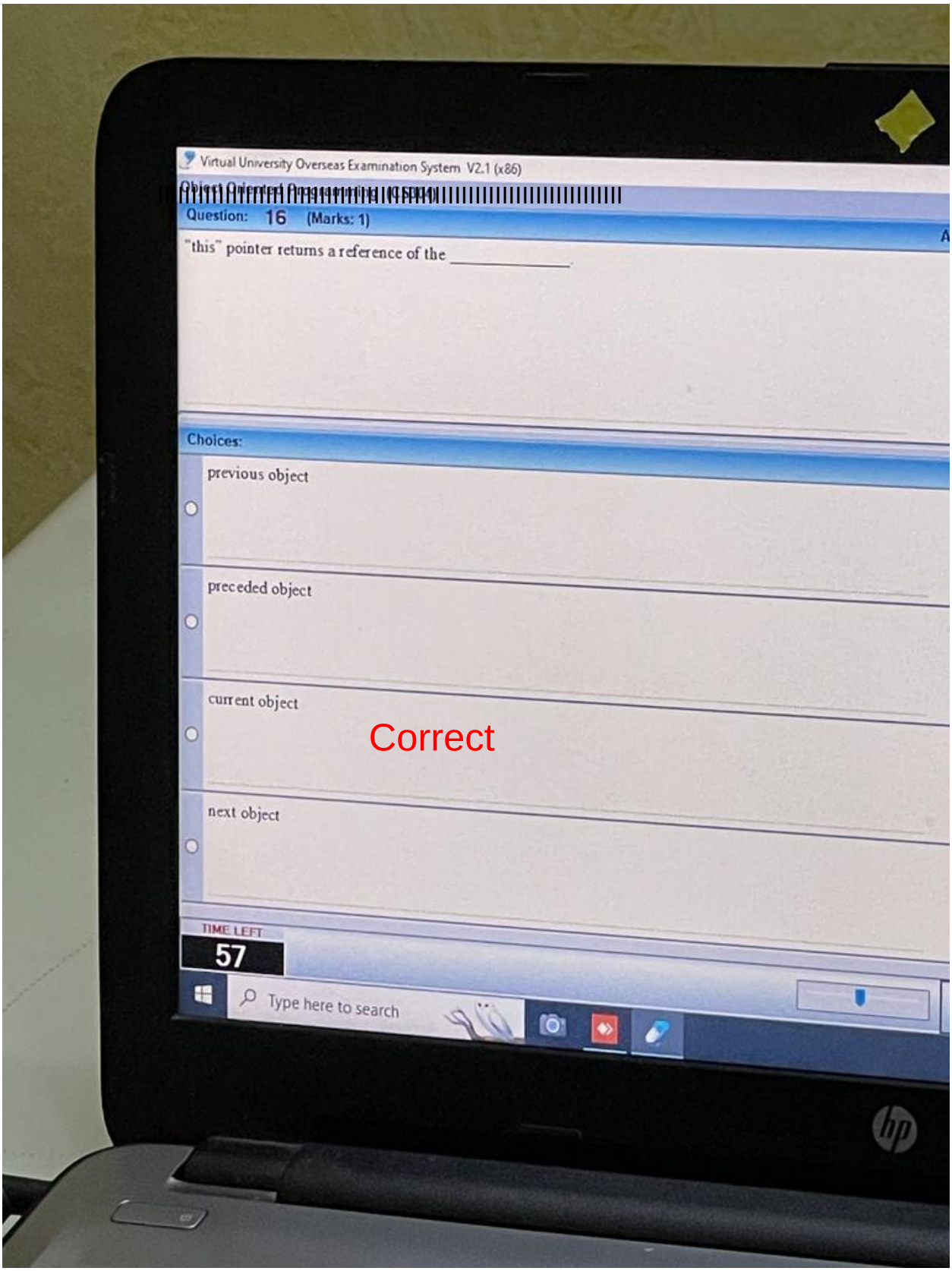
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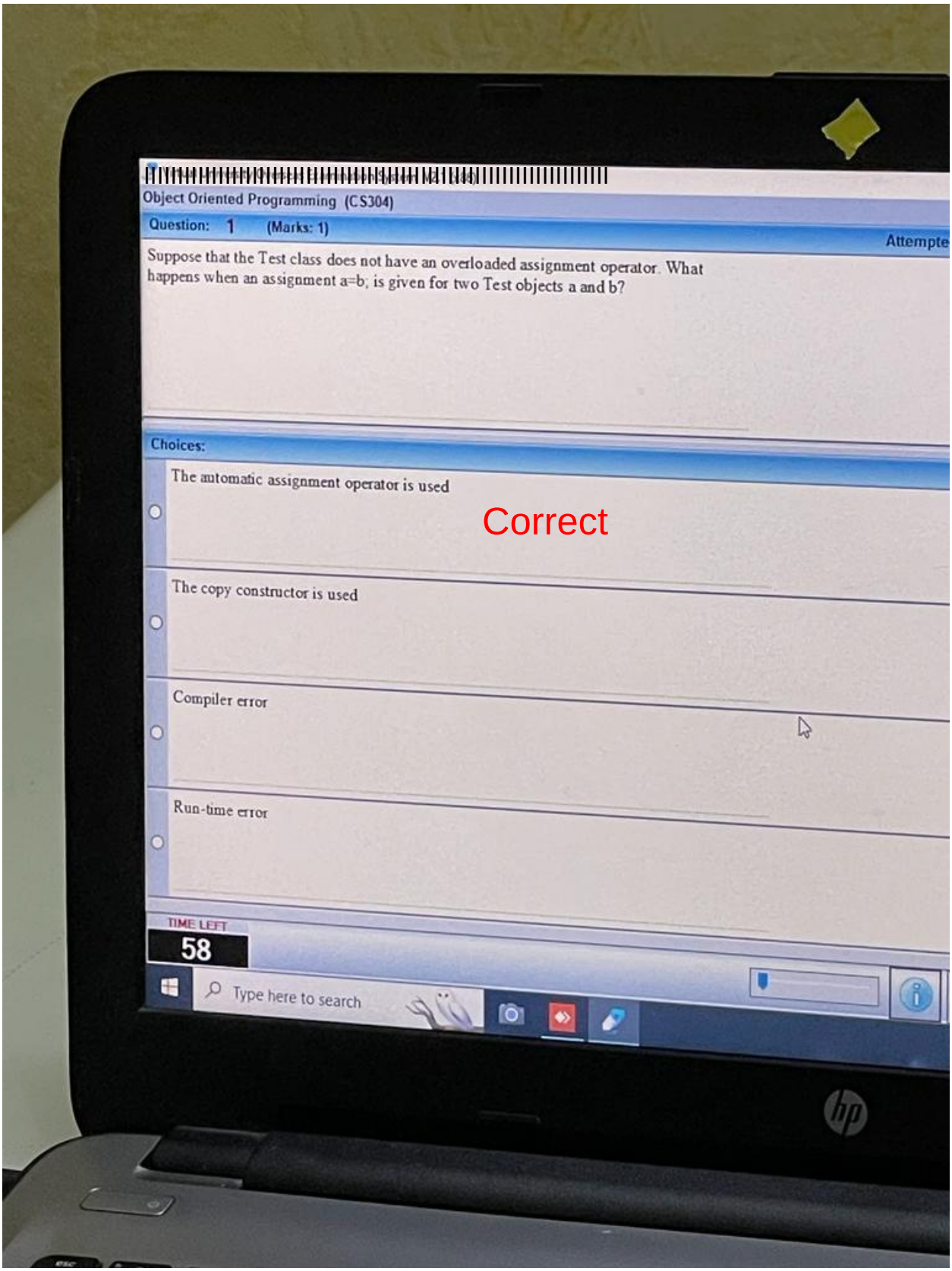
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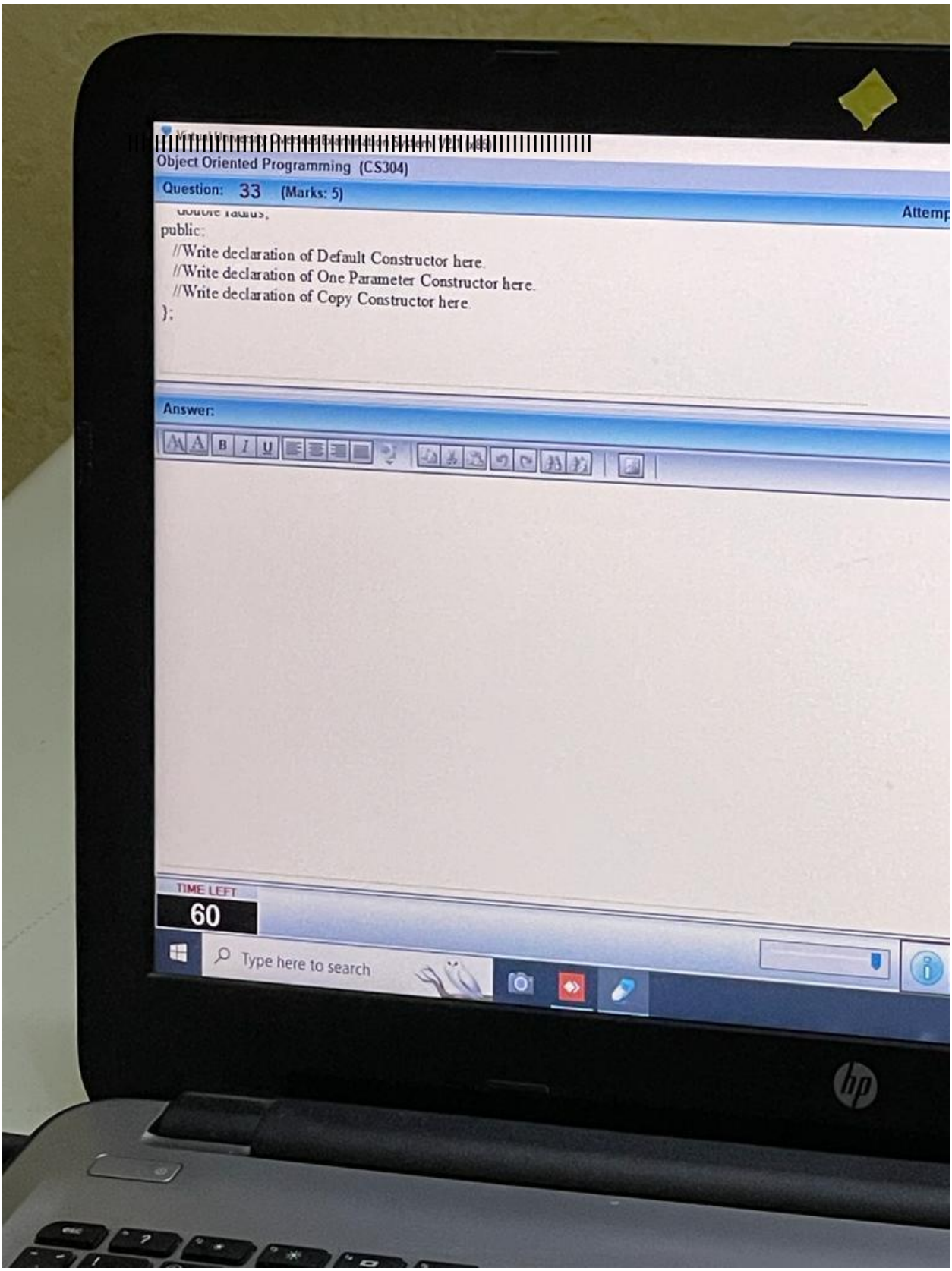
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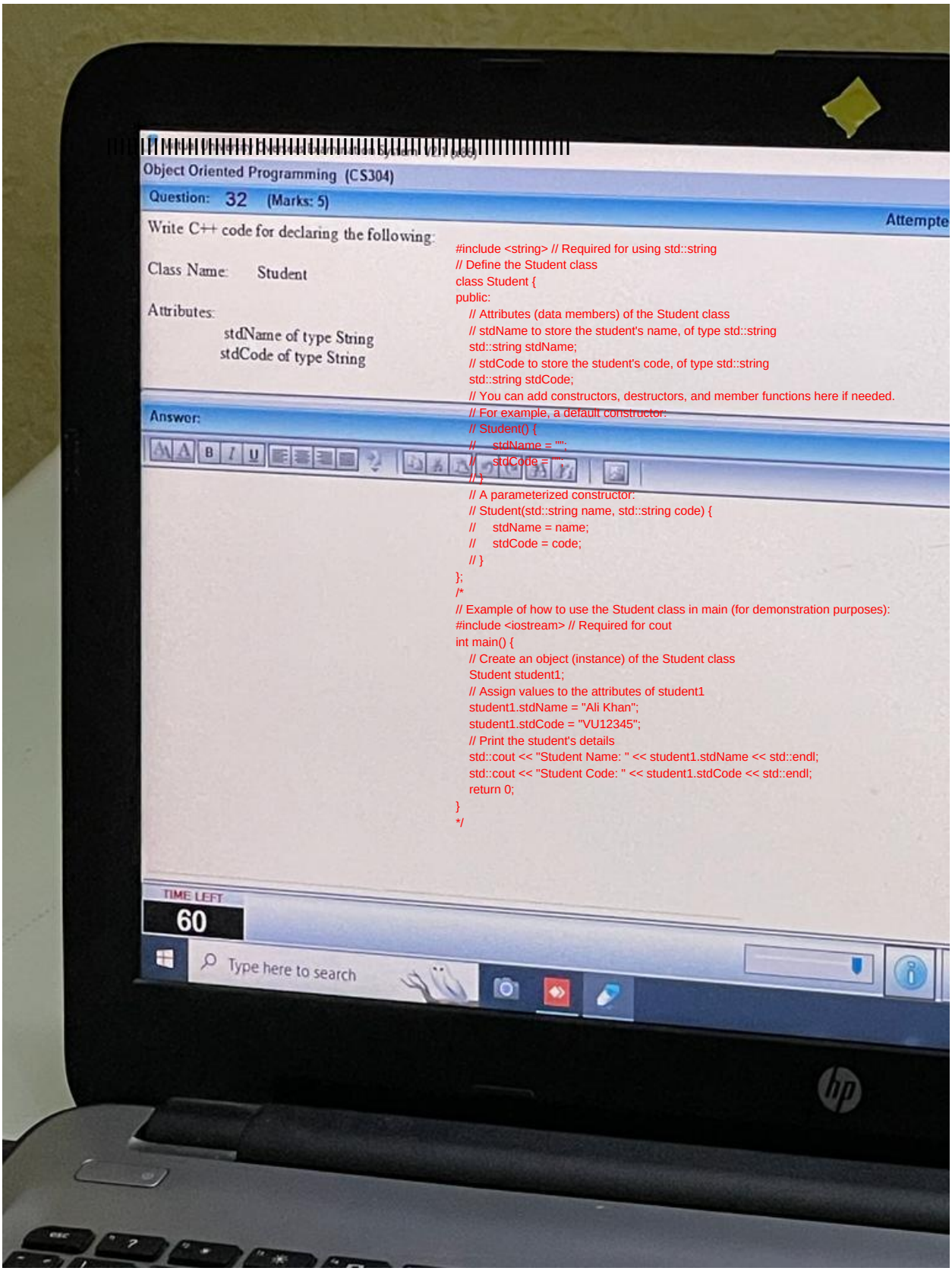
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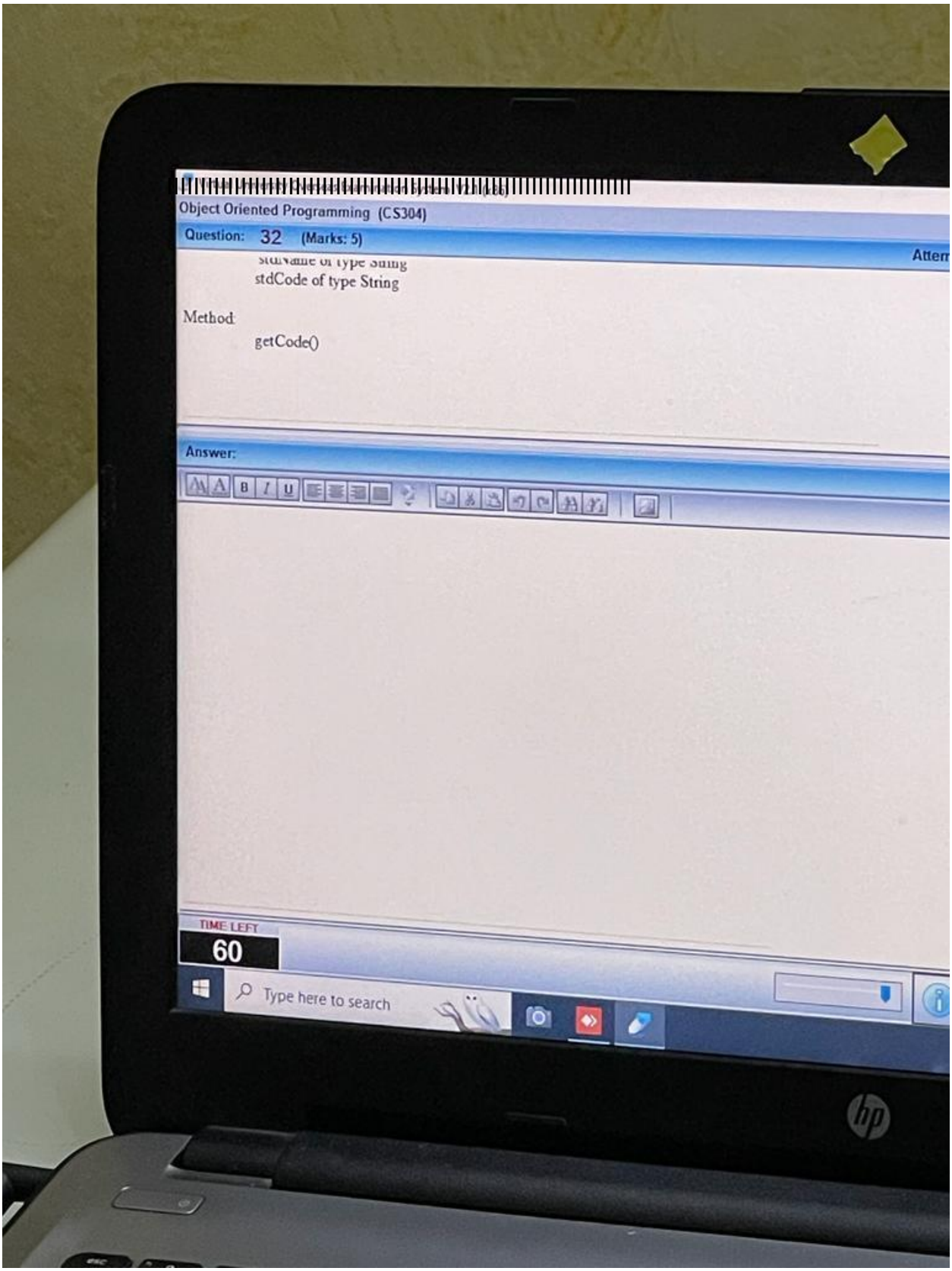
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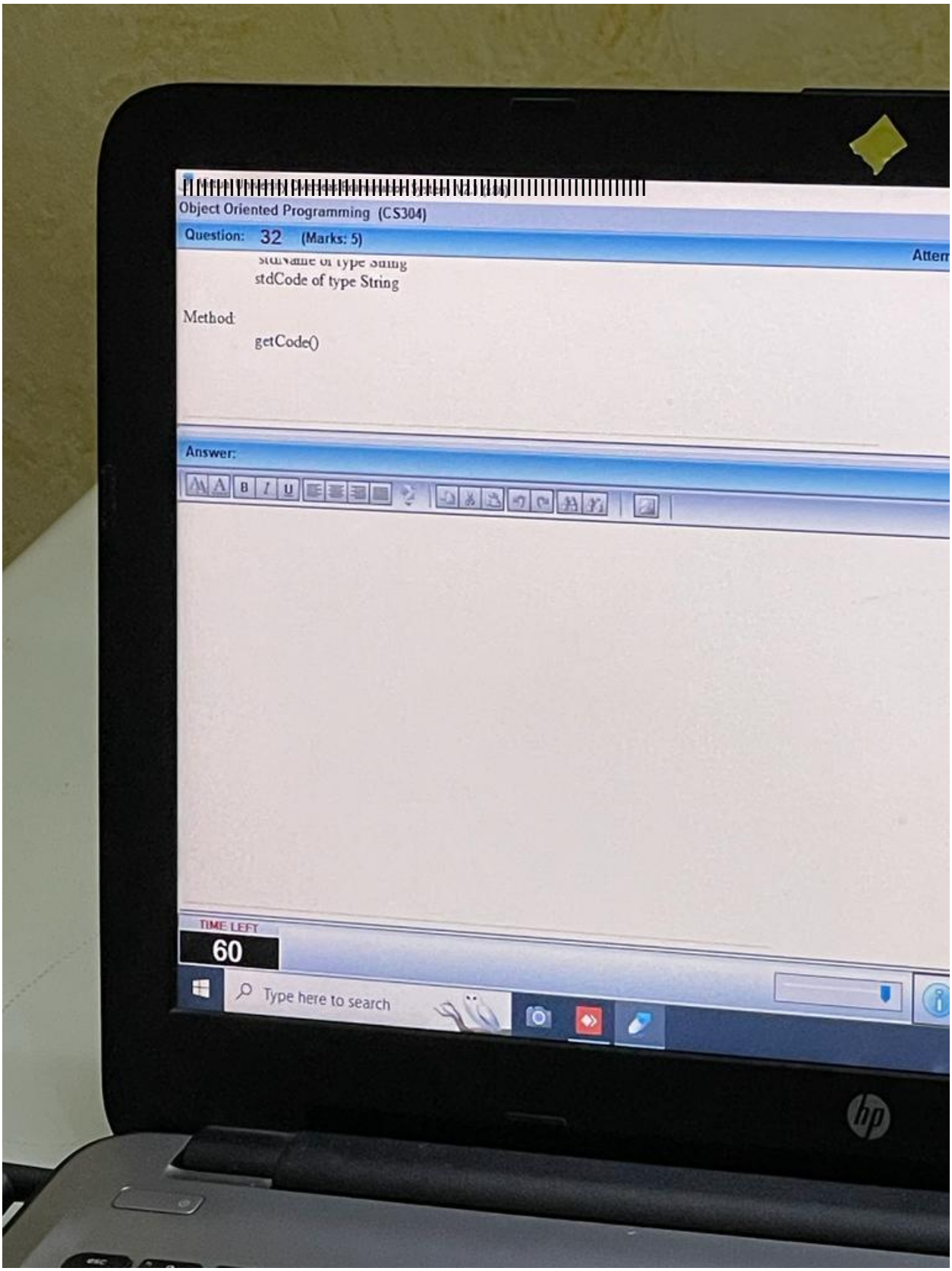
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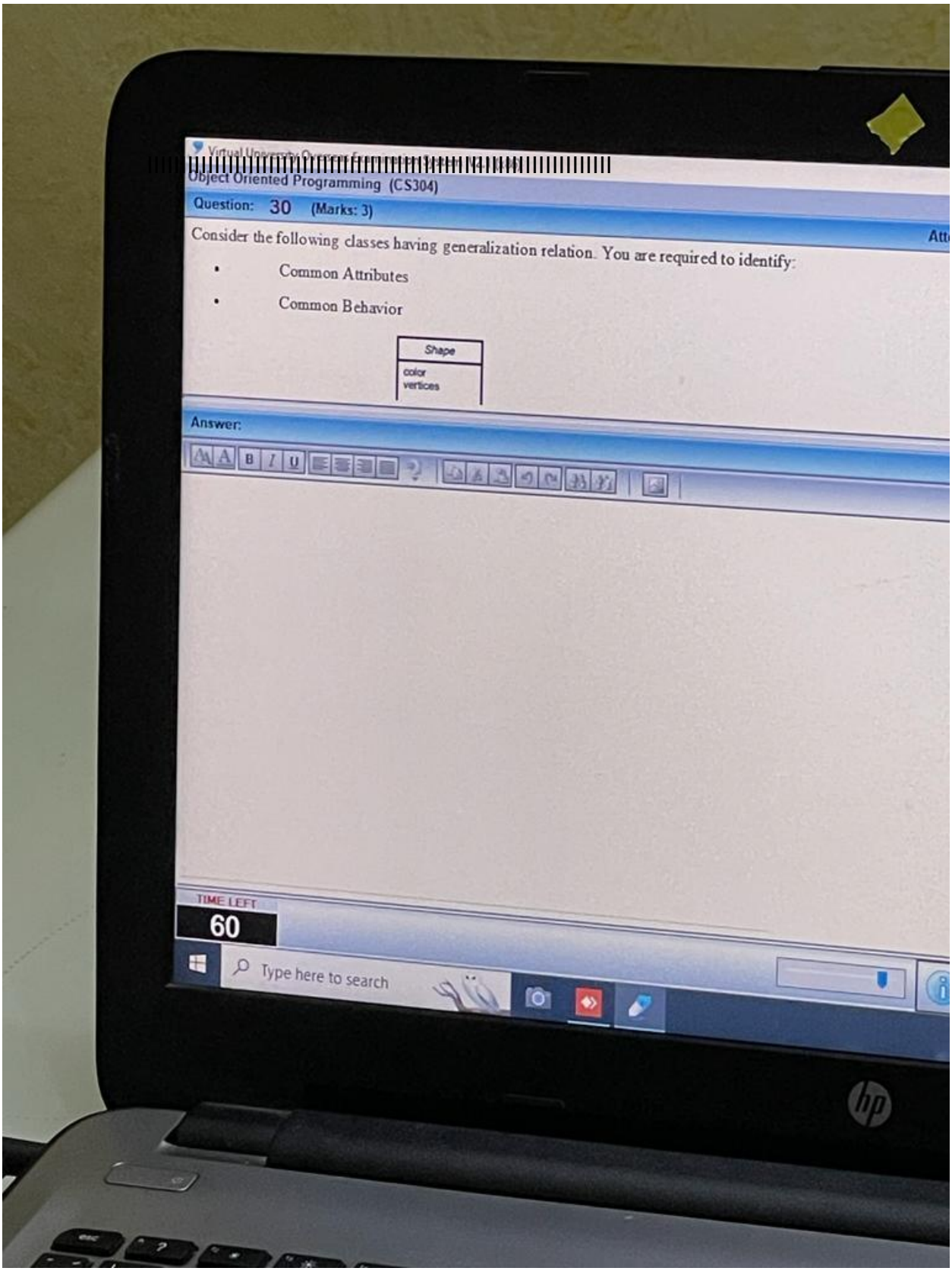
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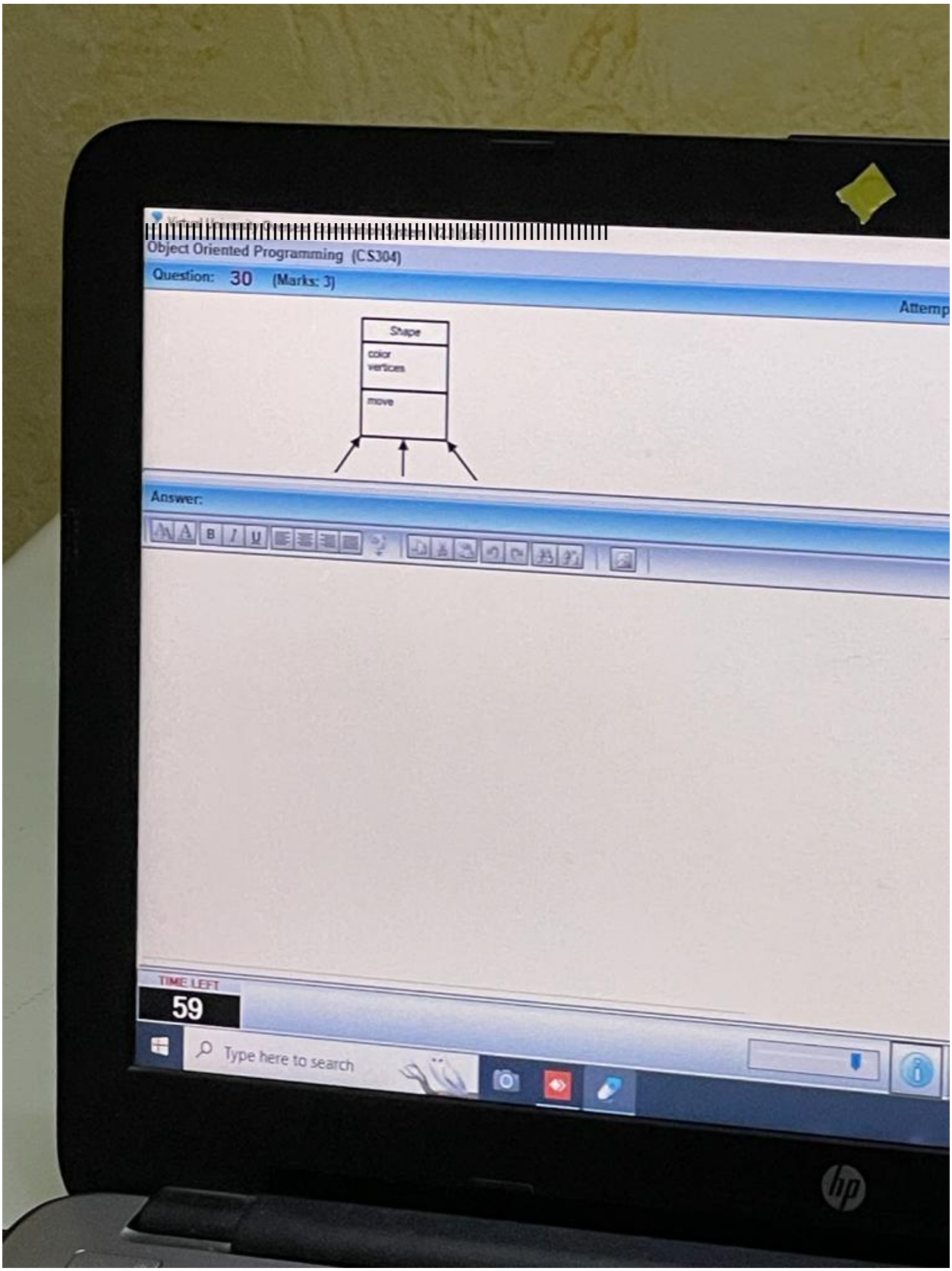
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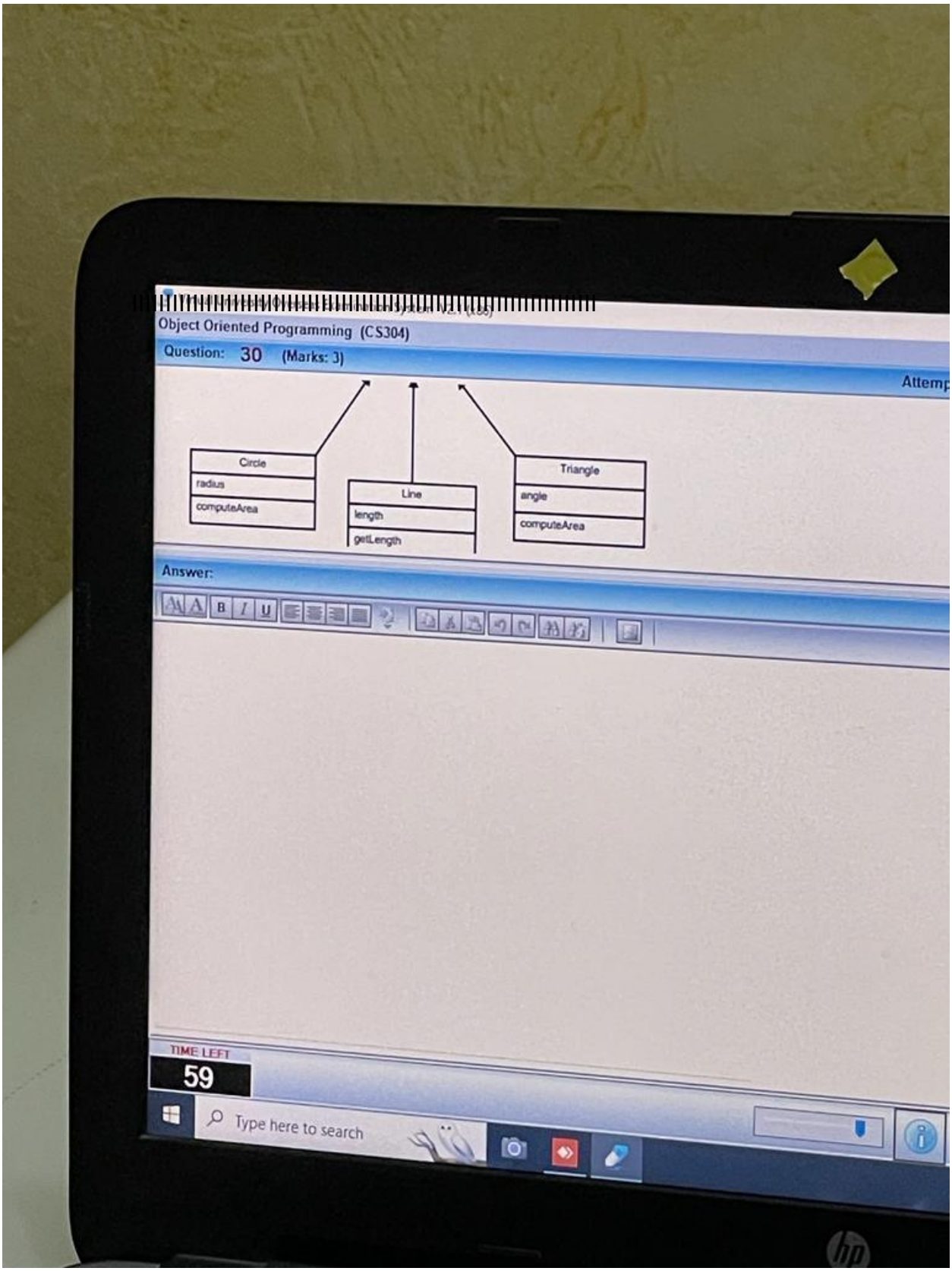
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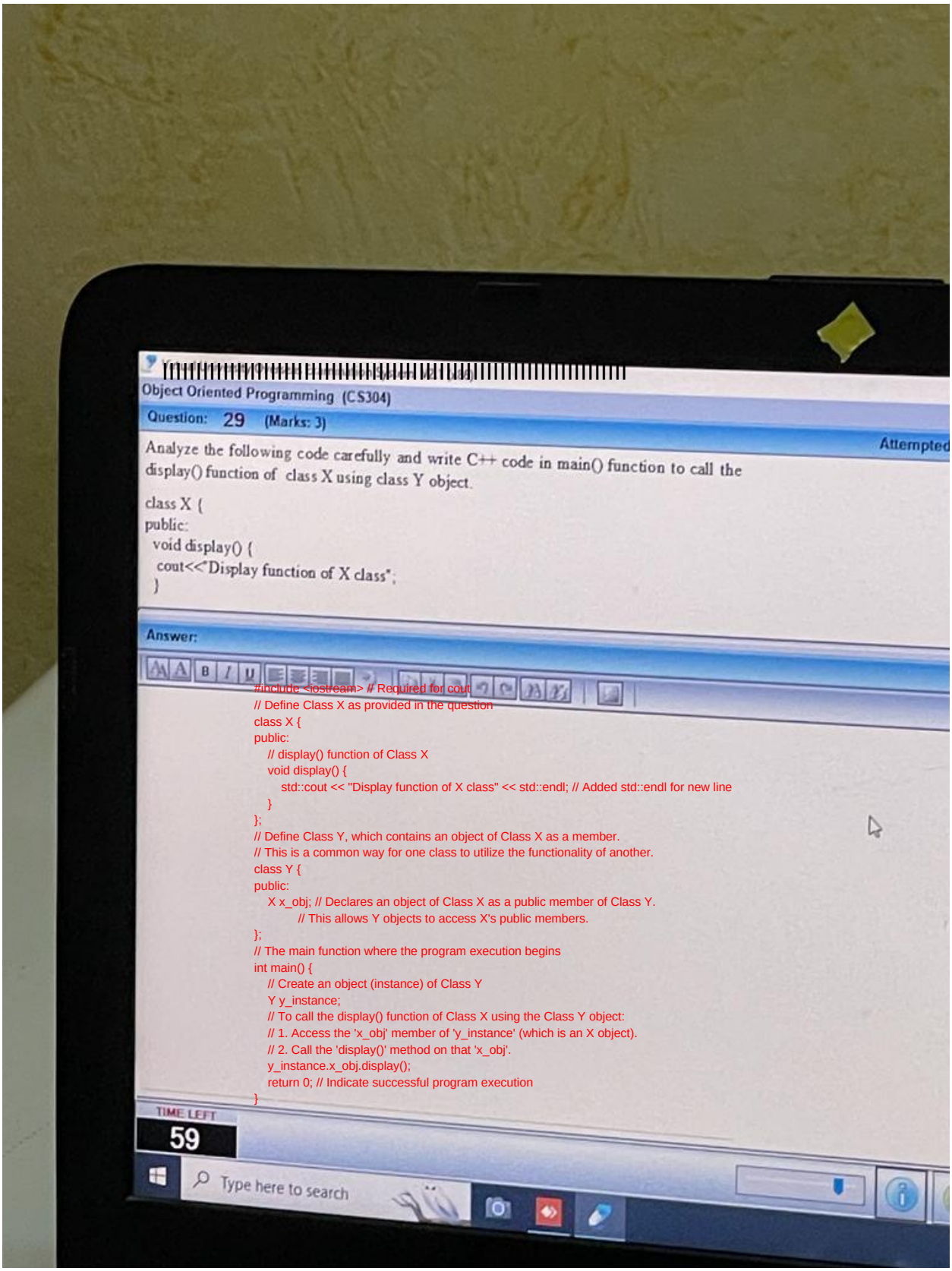
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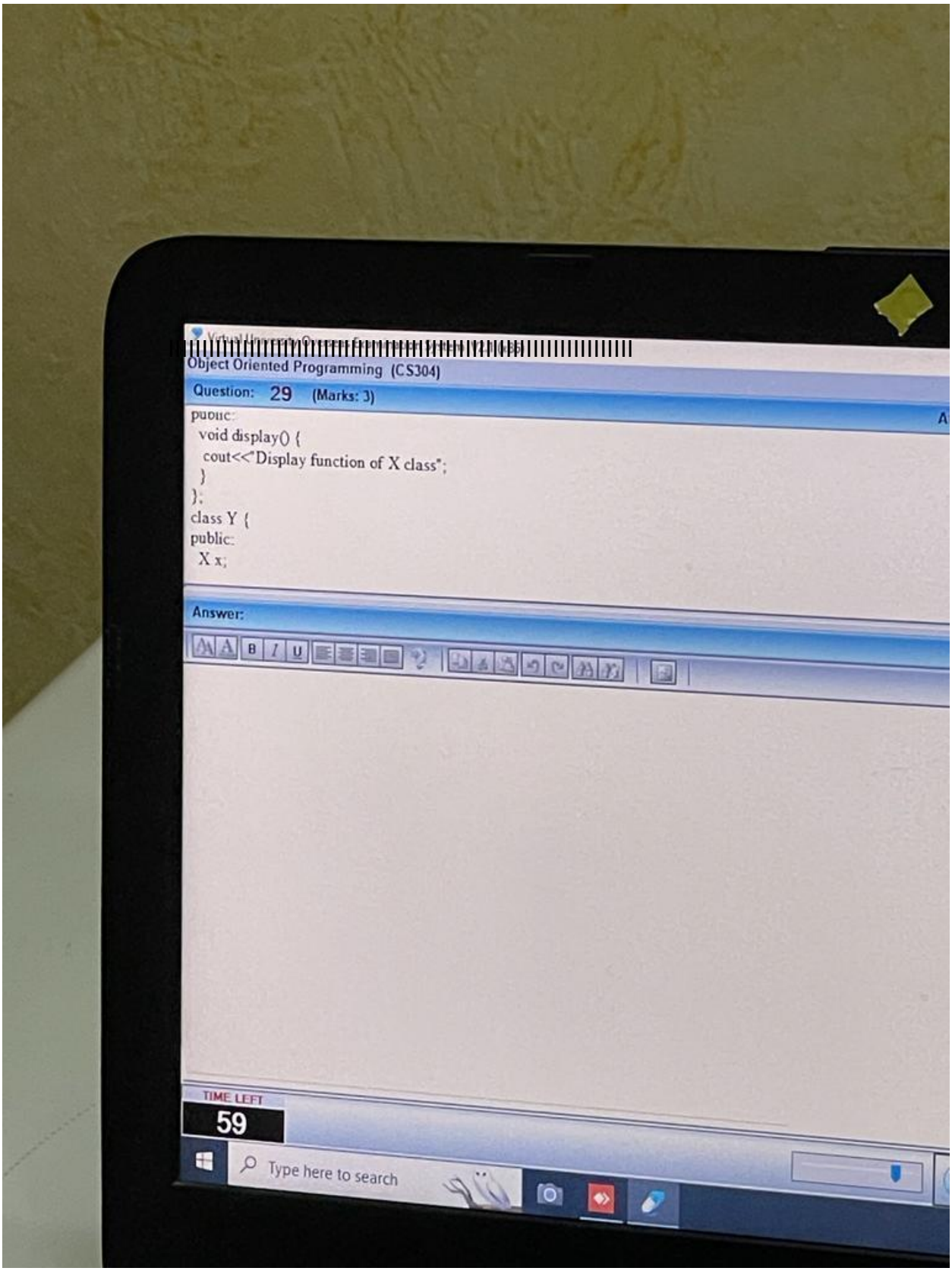
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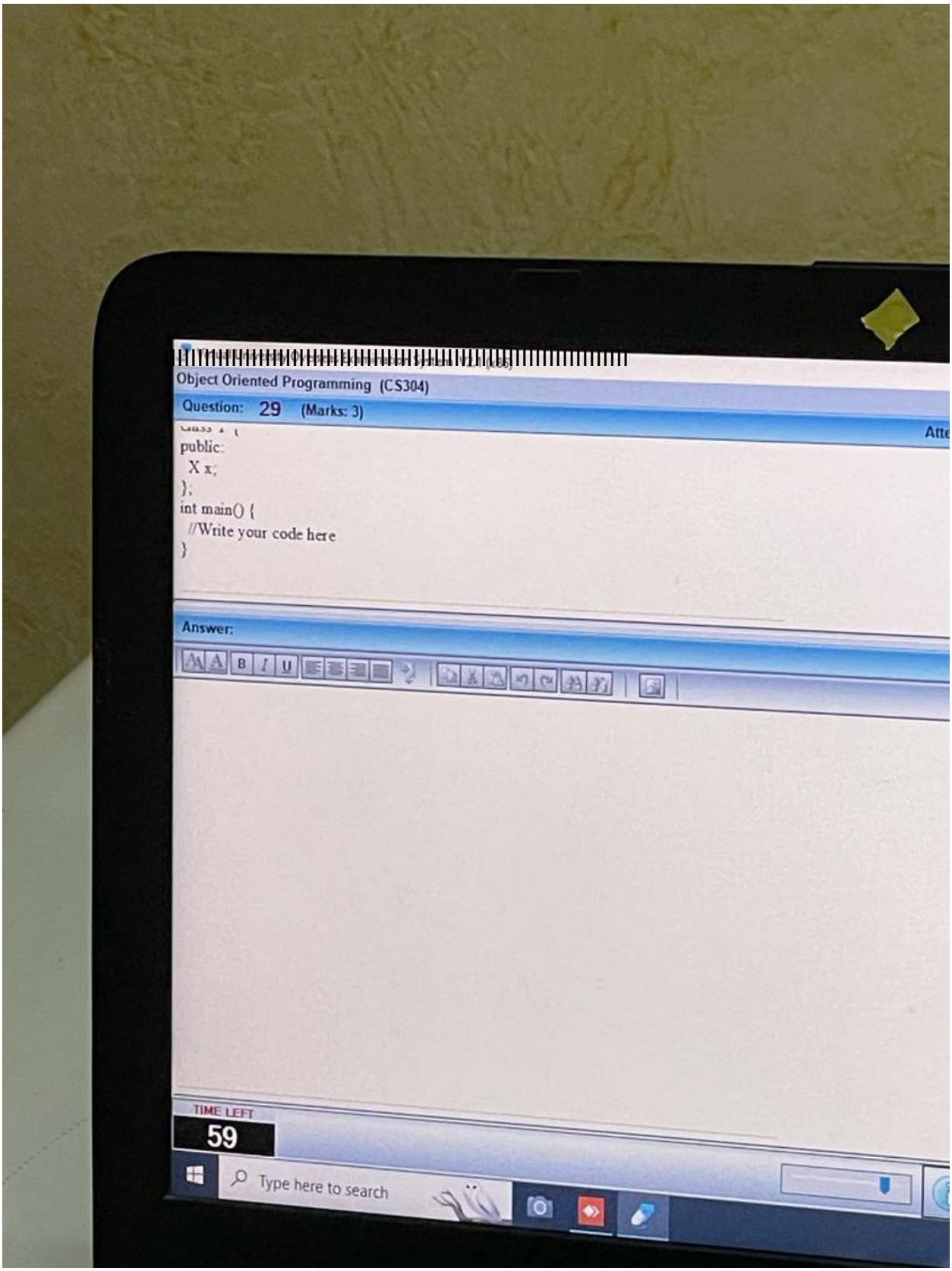
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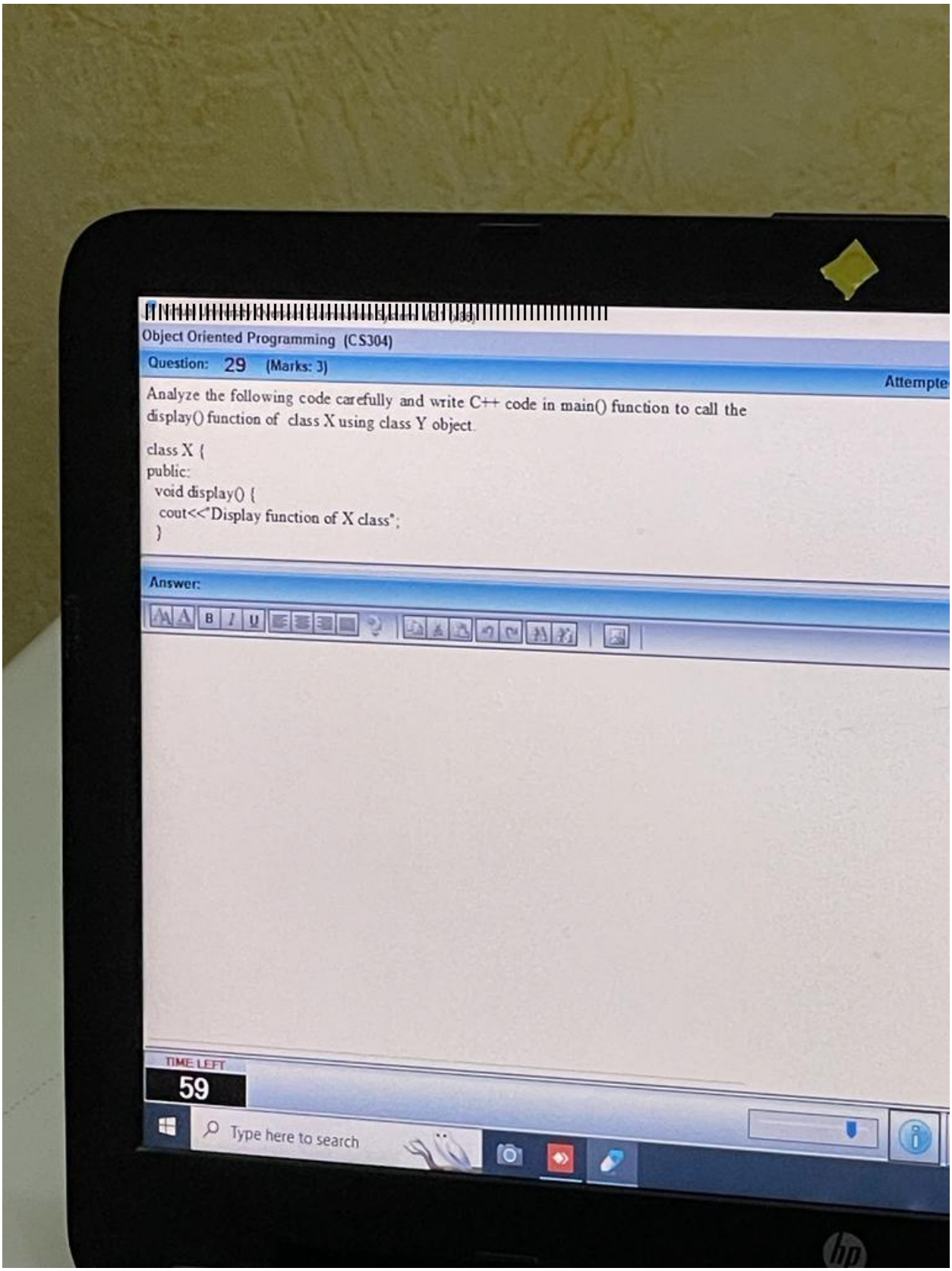
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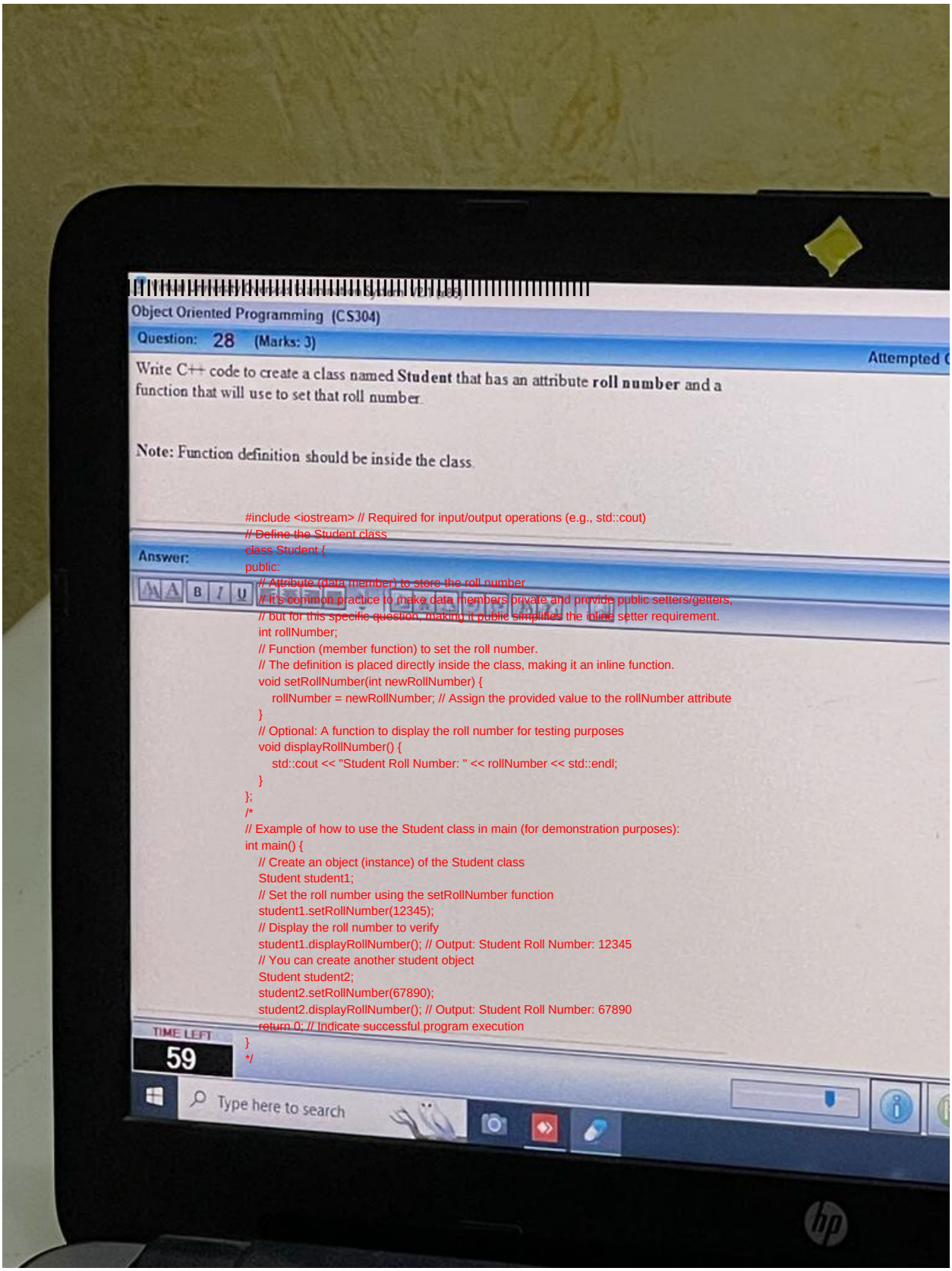
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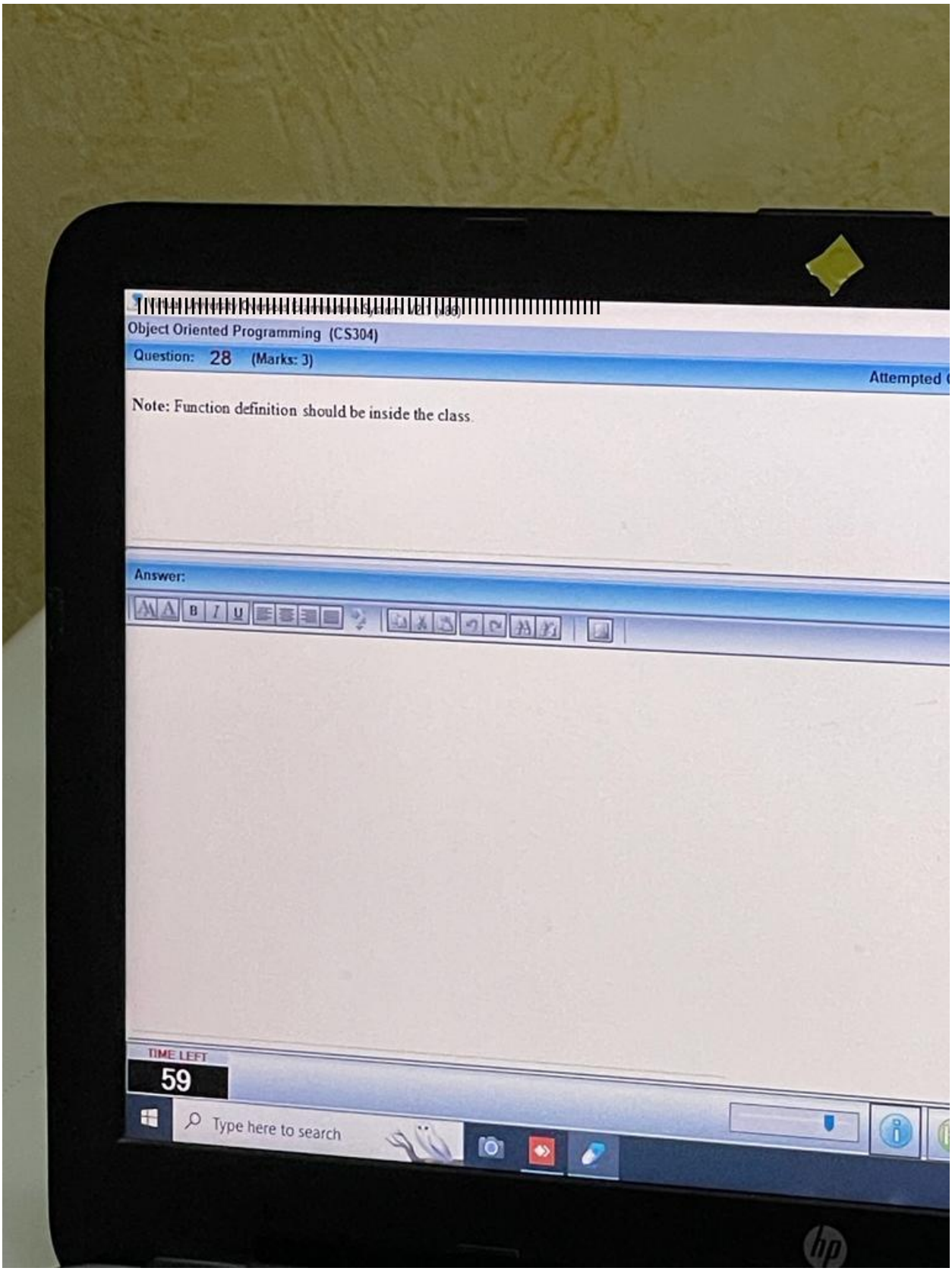
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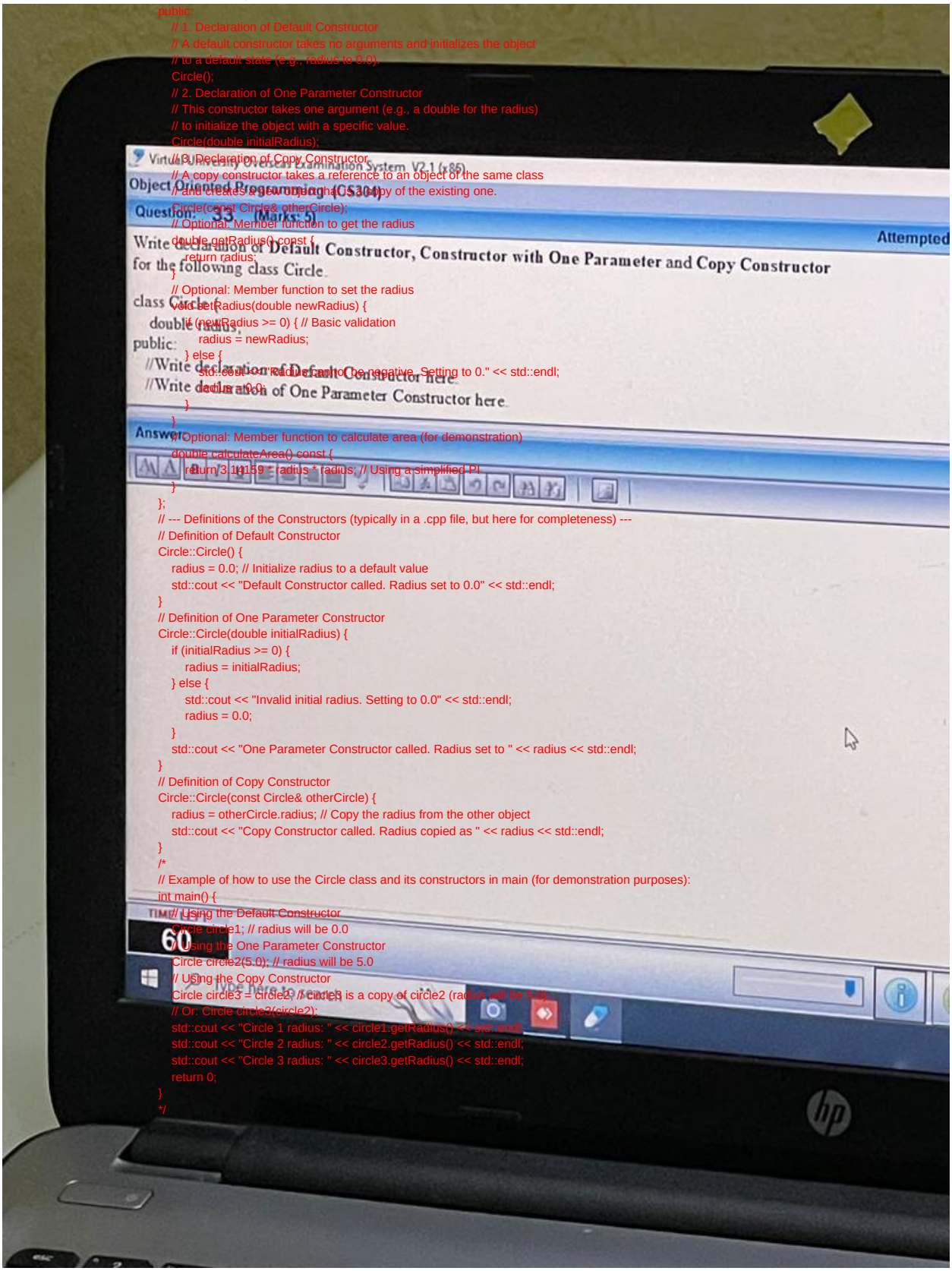
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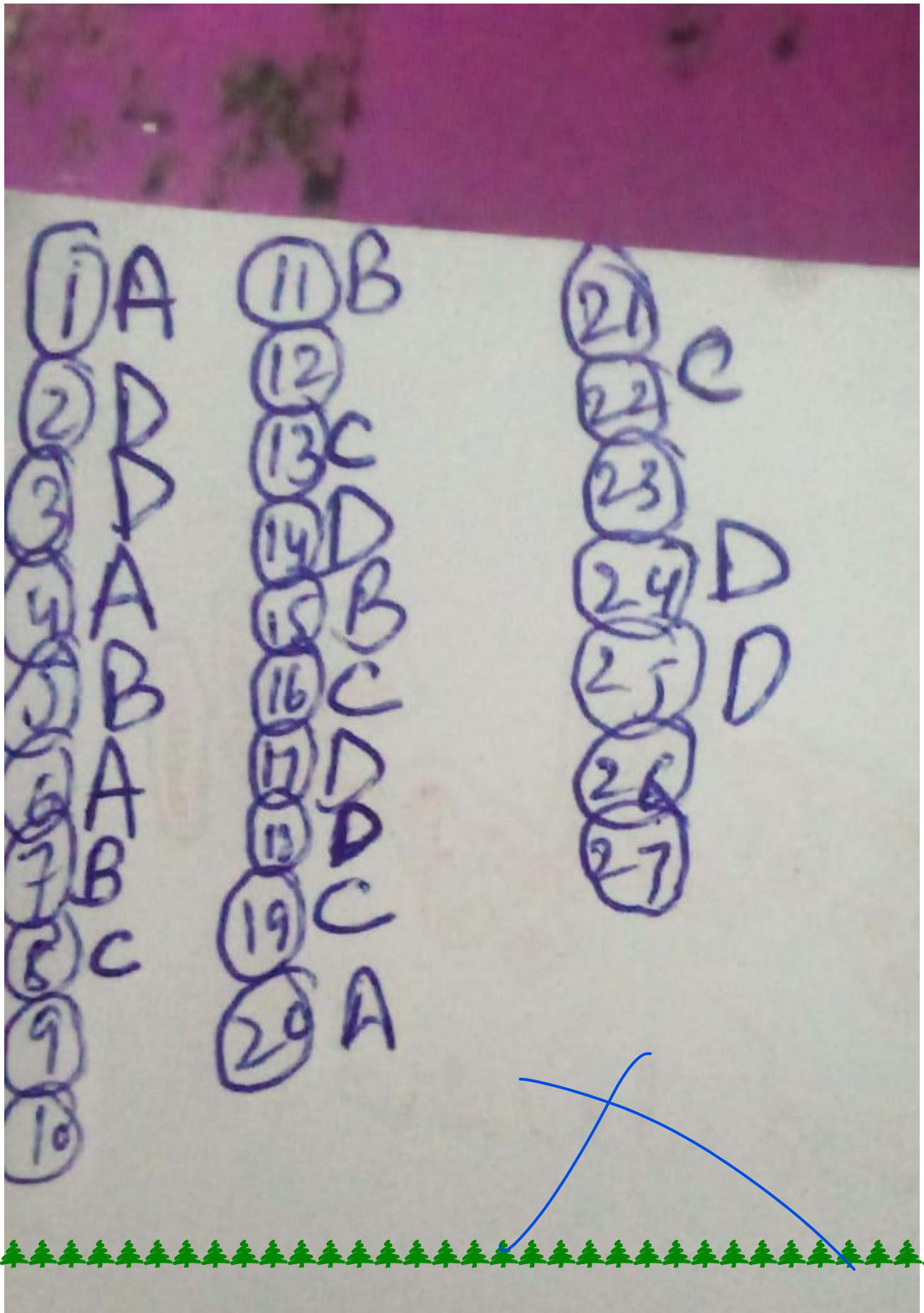



```
#include <iostream> // Required for input/output operations (e.g., std::cout)
// Define the Circle class
class Circle {
private: // It's good practice to make data members private
    double radius;
public:
    // 1. Declaration of Default Constructor
    // A default constructor takes no arguments and initializes the object
    // to a default state (e.g., radius to 0.0).
    Circle();
    // 2. Declaration of One Parameter Constructor
    // This constructor takes one argument (e.g., a double for the radius)
    // to initialize the object with a specific value.
    Circle(double initialRadius);
    // 3. Declaration of Copy Constructor
    // A copy constructor takes a reference to an object of the same class
    // and creates a new object as a copy of the existing one.
    Circle(const Circle& otherCircle);
    // Optional: Member function to get the radius
    double getRadius() const {
        return radius;
    }
    // Optional: Member function to set the radius
    void setRadius(double newRadius) {
        if (newRadius >= 0) { // Basic validation
            radius = newRadius;
        } else {
            std::cout << "Radius cannot be negative. Setting to 0." << std::endl;
            radius = 0.0;
        }
    }
    // Optional: Member function to calculate area (for demonstration)
    double calculateArea() const {
        return 3.14159 * radius * radius; // Using a simplified PI
    }
};
// --- Definitions of the Constructors (typically in a .cpp file, but here for completeness) ---
// Definition of Default Constructor
Circle::Circle() {
    radius = 0.0; // Initialize radius to a default value
    std::cout << "Default Constructor called. Radius set to 0.0" << std::endl;
}
// Definition of One Parameter Constructor
Circle::Circle(double initialRadius) {
    if (initialRadius >= 0) {
        radius = initialRadius;
    } else {
        std::cout << "Invalid initial radius. Setting to 0.0" << std::endl;
        radius = 0.0;
    }
    std::cout << "One Parameter Constructor called. Radius set to " << radius << std::endl;
}
// Definition of Copy Constructor
Circle::Circle(const Circle& otherCircle) {
    radius = otherCircle.radius; // Copy the radius from the other object
    std::cout << "Copy Constructor called. Radius copied as " << radius << std::endl;
}
/*
// Example of how to use the Circle class and its constructors in main (for demonstration purposes):
int main() {
    // Using the Default Constructor
    Circle circle1; // radius will be 0.0
    // Using the One Parameter Constructor
    Circle circle2(5.0); // radius will be 5.0
    // Using the Copy Constructor
    Circle circle3 = circle2; // circle3 is a copy of circle2 (radius will be 5.0)
    // Or: Circle circle3(circle2);
    std::cout << "Circle 1 radius: " << circle1.getRadius() << std::endl;
    std::cout << "Circle 2 radius: " << circle2.getRadius() << std::endl;
    std::cout << "Circle 3 radius: " << circle3.getRadius() << std::endl;
    return 0;
}
*/
```

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ANSWER

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Q30

Shape is common attributes. .
Color vertices are common attributes.
move is common behaviour

Q33

```
class Circle {
private:
    double radius;

public:
    // Default Constructor
    Circle();

    // Constructor with One Parameter
    Circle(double r);

    // Copy Constructor
    Circle(const Circle& otherCircle);
};
```

In the above declarations:

Q28

```
#include <iostream>

class Student {
private:
    int RollNumber; // Attribute

public:
    // Member function to set the roll number
    void setRollNumber(int roll) {
        RollNumber = roll;
    }

    // Member function to get the roll number (optional)
    int getRollNumber() {
        return RollNumber;
    }
};

int main() {
```

```
    // Creating an instance of the Student class
    Student student1;

    // Setting the roll number using the setRollNumber function
    student1.setRollNumber(12345);

    // Getting the roll number using the getRollNumber function (optional)
    std::cout << "Roll Number: " << student1.getRollNumber() << std::endl;

    return 0;
}
```

Q32.

```
cs304 mid term by iqra azhar#include <iostream>
#include <string>

class Student {
private:
    std::string StdName;
    std::string StdCode;

public:
    // Constructor
    Student(std::string name, std::string code) : StdName(name), StdCode(code) {}

    // Method to get the student code
    std::string getcode() {
        return StdCode;
    }
};

int main() {
    // Creating an instance of the Student class
    Student student1("John Doe", "S12345");

    // Calling the getcode() method
    std::cout << "Student Code: " << student1.getcode() << std::endl;

    return 0;
}
```

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